

Help Me Do It on My Own: Sensory Data Analysis with jamovi and the SEDA Module

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Preface

This booklet was born from a simple conviction: we learn better to do things on our own when someone has taken the time to explain not only **how**, but also **why**.

The statistical analysis of sensory data has long been learned in R, through books, scripts, and tutorials that assumed a certain familiarity with programming, packages, and sometimes even a degree of resilience in the face of error messages. These tools have their strengths, their elegance, and remain indispensable. But they can also be a barrier for many practitioners who nevertheless work regularly with sensory data, without coming from the world of statistics or coding.

jamovi has partly changed the situation. By making methods accessible through its interface that were once reserved for more technical users, it has widened access to the statistical analysis of sensory data. But making a method clickable is not enough to make it understandable. A button without explanation remains a black box; a result without a reading framework remains fragile; a map without interpretation remains mute.

This is precisely the space this booklet seeks to occupy.

It replaces neither a full statistics course nor in-depth training in R. Its ambition is more focused: to give you, for each analysis, **enough understanding to work with discernment**. Enough to know what a confidence ellipse really represents. Enough to understand why a product may be absent from a descriptive table without thereby being “without a profile”. Enough not to read a preference map as an absolute truth far from the products actually tested. Enough, above all, no longer to be entirely dependent on an interface that one uses without understanding.

The guiding thread of this book is therefore **autonomy**. Not the immediate autonomy of someone who already knows everything, but the gradual autonomy of someone who can move forward with reliable points of reference, connect the visible options in jamovi to their statistical logic, and interpret the outputs more accurately. In that sense, this booklet deliberately sits between two worlds: that of concrete practice in jamovi, and that of methodological understanding, which gives meaning to the results.

This booklet covers the **seven analyses** of the **SEDA** module, in the following order: **univariate characterisation of the product space**, **multivariate representation with confidence ellipses**, **CATA** data analysis, **JAR** data analysis, **Napping**, **free sorting**, and finally **preference mapping**. Together, they form a coherent progression: from fixed-list attribute methods to more holistic approaches, and then to hedonic analyses. Each method

corresponds to a particular way of producing data, asking a question, and interpreting a result. You can easily try these methods using the datasets distributed with the **SEDA** module, making it easier to become familiar with the graphs and outputs generated by each analysis.

To maintain overall coherence, each chapter follows the same structure: an introduction that sets out the statistical problem, a section explaining what is really happening behind the interface, a detailed presentation of the options, a step-by-step procedure, and then keys for interpreting the tables and graphs. This choice is not merely editorial; it is pedagogical. It is intended to ensure that, from one chapter to the next, the reader finds stable reference points and can more easily transfer what has been learned.

This booklet is intended for sensory evaluation practitioners, engineers, students, teachers, researchers discovering jamovi, and more broadly for anyone who needs support between a course and a real field analysis.

Each chapter can be read independently. You can enter this booklet through the method that corresponds to the data you currently have. But the whole has also been conceived as a journey: the further one goes, the more one understands that sensory analysis is not a stack of separate techniques, but a family of complementary approaches, articulated through common logics of representation, description, and interpretation.

At heart, this book pursues a very simple idea: a good tool should not merely make it possible to obtain a result, but should help one understand what that result means. If these pages allow you to move from “*show me how to do it*” to “*now I can do it on my own*”, then they will have fully achieved their purpose.

1 Introduction

1.1 Presenting jamovi

1.1.1 What is jamovi?

In one sentence: *jamovi is a free, modern, and accessible statistics package, designed for both teaching and research.*

Developed by contributors from the JASP project, jamovi relies on R for its statistical calculations while providing a graphical interface that requires no code at all. In practical terms, its interface works like a dynamic spreadsheet¹: the data are visible and editable in a sheet on the left, and the results update in real time on the right. This organisation makes the tool accessible to both beginners and experienced users.

One often underestimated advantage of jamovi is its ability to save everything in a single `.omv` file: the data, the analysis options selected, and the results. This greatly facilitates reproducibility and the sharing of work, two areas in which traditional spreadsheets show their limits.

A comparison with other common tools is useful for situating jamovi:

Feature	Excel / Google Sheets	jamovi
Data entry	Yes	Yes
Statistical analyses	No (requires external tools)	Yes (built in natively)
Reproducibility	Difficult to track	Yes, via the <code>.omv</code> format
Advanced modules	No	Yes, via extensions
Open source	No	Yes

What about XLSTAT? XLSTAT is a paid-for Excel plug-in that adds advanced statistical functions. Although it is powerful, it remains proprietary, dependent on Microsoft Excel, and its cost may be a barrier for students or organisations with limited budgets. jamovi offers comparable features, and in some areas superior ones, while being entirely free and independent.

¹A *spreadsheet* is a data-processing tool organised into rows and columns, such as Microsoft Excel or Google Sheets.

1.1.2 How do you install jamovi?

Installation is straightforward and requires no technical skills. jamovi is cross-platform: it works in the same way on Windows, macOS, and Linux. To install it, go to jamovi.org/download, choose your operating system, then run the downloaded file.

On the download page, two versions are offered. The **Solid** version is the most stable and thoroughly tested: it is the one to favour for teaching, practical classes, and research. The **Current** version includes the very latest features but may contain a few minor instabilities.

1.2 Modules in jamovi

In jamovi, a module is the equivalent of a package in R or Python: it adds specific features to the software, whether in the form of new types of analyses, graphs, or statistical models.

The jamovi module ecosystem is already rich and covers many needs:

Module	Main features
GAMLj	Linear, logistic, and hierarchical models
semLj	Structural equation models
jsq	Bayesian methods
PAMLj	Statistical power analysis
lsj	Interactive exercises from the <i>Learning Stats</i> manual
MEDA	PCA, CA, MCA, MFA (FactoMineR under jamovi)
SEDA	Sensory data analysis (SensomineR under jamovi)

To install a module, simply open jamovi, go to the **Modules** tab in the top right-hand corner, select **jamovi library**, then choose the module you want and click **Install**. Some modules still under development may also be installed manually from a `.jmo` file.

1.3 The SEDA module

SEDA (*Sensory Evaluation Data Analysis*) is the jamovi module dedicated to the statistical analysis of sensory data. It is based on the R package *SensomineR* and makes the main statistical methods used in sensory analysis available through a graphical interface — without requiring a single line of code.

Its menu is organised into three sections corresponding to three broad ways of producing sensory data.

The first section, **Fixed List of Attributes**, brings together methods in which subjects evaluate stimuli using a list of attributes defined in advance. This is the classic case of QDA (*Quantitative Descriptive Analysis*) data, where a trained panel scores the intensity of each attribute for each product. Here, SEDA offers two complementary analyses: one to **characterise the stimuli attribute by attribute** (by identifying which attributes differentiate them and which characterise each stimulus in particular), and another to **represent the sensory space in a multivariate way** (by producing a map of the stimuli enriched with confidence ellipses reflecting the stability of the positions according to panel composition). This section also includes the analysis of **CATA** (*Check-All-That-Apply*) data, in which respondents tick from a list the terms they feel are present in a stimulus, and the analysis of **JAR** (*Just About Right*) data, in which respondents indicate whether each attribute is too strong, too weak, or just about right — making it possible to estimate the *penalties* these deviations impose on liking.

The second section brings together **holistic approaches**, in which subjects evaluate the stimuli globally, without any predefined list of attributes. Two methods are covered: **Napping** (*projective mapping*), in which each subject positions the stimuli on a rectangle according to their perceived similarities, and **free sorting** (*sorting task*), in which each subject freely groups the stimuli into categories. In both cases, SEDA produces a consensual representation of the stimuli by combining the individual maps — by Multiple Factor Analysis for Napping, and by Multiple Correspondence Analysis for free sorting — and may complement this representation with an automatic classification of the stimuli.

The third section, **Hedonic Data**, is dedicated to **liking data**. It contains **preference mapping**, which combines an already constructed representation space (for example, from a PCA) with individual liking scores in order to identify favourable regions of that space. SEDA can also classify consumers from their centred liking profiles and produce preference maps for the resulting segments. This analysis is the natural extension of the previous analyses: once the sensory space has been constructed and interpreted, preference mapping helps identify which regions are favourable and whether the same preference structure is shared across consumer segments.

This booklet covers all seven of these analyses, in the order of the SEDA menu. Throughout the rest of this booklet, the terms *product* and *stimulus* mean the same thing and are used interchangeably.

1.4 From jamovi to R: the Rj Editor

SEDA can display the R script corresponding to the options selected in the interface. Throughout this booklet, these scripts provide a bridge from a point-and-click analysis to a reproducible R workflow.

Before using the **Rj Editor** for the first time, open its settings with the gear icon and select **System R** rather than **jamovi R**. Your system R installation must contain the packages required by the analysis — mainly **SensoMineR** and **FactoMineR** in this booklet — as well as **jmvconnect**, which gives the Rj Editor access to the dataset currently open in jamovi under the name **data**.

When an analysis offers the **R code** option, check it to display a script that can be copied into the Rj Editor. The chapters below explain the main functions and arguments used by each generated script.

1.5 Some useful references

- Official website: jamovi.org
- [Online documentation](#)
- [jamovi user manual](#)
- Navarro, D., Foxcroft, D., & Meunier, J.-M. (2020). *Apprentissage des statistiques avec Jamovi*.
- Love, J., Selker, R., Marsman, M. et al. (2019). *JASP: Graphical statistical software for common statistical designs*. [Journal of Statistical Software](#), 88, 1–17.
- Navarro, D., & Foxcroft, D. (2018). *Learning statistics with jamovi*.
- Community and modules: forum.jamovi.org

2 QDA Data — Characterisation of the Sensory Space

2.1 Introduction

This analysis is designed for **QDA** (*Quantitative Descriptive Analysis*) data, when several subjects evaluate several stimuli using a fixed list of sensory attributes. It answers two complementary questions: **which attributes structure the sensory space as a whole** and **which attributes characterise each stimulus in particular**. More specifically, the function returns two families of results: a ranked list of the attributes that differentiate the set of stimuli, then, for each stimulus, a list of the attributes that characterise it.

Why not use a standard alternative?

A simple comparison of means by stimulus can be misleading if subjects do not all use the rating scale in the same way. Here, the analysis takes into account both the product effect (**Stimulus Effect**) and the subject effect (**Subject Effect**), which makes it possible to identify discriminating attributes and profiles more *robustly* than with a purely descriptive reading.

To test this function, simply load the *sensochoc* data available with the SEDA module — they appear in the list of example datasets when opening a new file in jamovi.

2.2 Understanding what happens under the hood

2.2.1 Data checking and preparation

The analysis first checks that the variable placed in the **Stimulus Effect** field and the variable placed in the **Subject Effect** field can be treated as factors, and that all variables placed in **Sensory Attributes** are numeric. If a sensory attribute is not numeric, the analysis stops with an error message.

2.2.2 An attribute-by-attribute analysis

A simple way to understand the computation is to say that the function proceeds **attribute by attribute** using an analysis of variance (ANOVA). For each variable in **Sensory Attributes**, it first evaluates whether the *product* effect is significant, while taking the *subject* effect into account. This first step acts as a global filter: it determines which attributes truly structure the sensory space.

2.2.3 Selection of discriminating attributes

Attributes whose significance is below the threshold defined by **Significance threshold (%)** are retained in the first results table. The default threshold is **5%**, which makes it a good starting point for a cautious reading. If no attribute passes this threshold, the function stops and indicates that no sensory attribute is discriminating at the selected threshold.

2.2.4 Stimulus-by-stimulus description

Once the globally discriminating attributes have been identified, the function examines, for each stimulus, which attributes significantly characterise it. Each stimulus is compared with a global reference, which makes it possible to identify attributes positively or negatively associated with that stimulus. The global reference corresponds to the mean of the stimuli on each attribute, that is, the average stimulus. The result takes the form of a second table grouped by stimulus.

2.2.5 The role of adjusted means

The **Adjusted mean** column can be understood as an estimate of the stimulus mean on an attribute, corrected for the subject effect. This column makes it possible to read the perceived level of a stimulus without being limited to a raw mean that may be influenced by subjects who are more “generous” or more “severe” in their ratings.

When the tasting design is complete, that is, when each subject evaluates all stimuli, the adjusted mean is equal to the arithmetic mean. It differs from it when data are missing; in that case, the adjusted mean is more reliable.

2.2.6 What it means when a stimulus is absent from the second table, Sensory Description of the Stimuli

A stimulus may not appear in the second table if no attribute significantly characterises it at the chosen threshold. This does not mean that it has “no profile”, but rather that it does not differ clearly enough from the mean of the full set of stimuli on the tested attributes.

2.3 Expected data structure

The data must contain:

- one column identifying the **stimuli**;
- one column identifying the **subjects**;
- several numeric columns corresponding to the **sensory attributes**.

Each row corresponds to the evaluation of one stimulus by one subject. If subjects evaluate the stimuli over several repetitions or several sessions, each evaluation must appear on a separate row.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variable intended for **Subject Effect** is categorical;
- all variables intended for **Sensory Attributes** are numeric.

2.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Characterization of the Stimulus Space**. The interface includes three variable drop zones, an **Instructions and R Code** section containing the **Read me before running** and **R code** checkboxes, and a numeric field, **Significance threshold (%)**.

2.4.1 Variables to complete

2.4.1.1 Stimulus Effect

Role: identifies the stimuli/products to compare. **Impact on the computation:** this variable defines the groups studied in the global analysis and in the stimulus-by-stimulus description. **Practical tip:** use explicit labels, because they directly structure the reading of the second table.

2.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is used to take systematic differences between subjects into account when estimating stimulus effects. **Practical tip:** check that the same subject keeps exactly the same identifier throughout the file.

2.4.1.3 Sensory Attributes

Role: groups together the sensory attributes to be analysed. **Impact on the computation:** the function applies the same analysis logic to each of these attributes; only numeric attributes are accepted. **Practical tip:** include all relevant attributes from your QDA; the statistical selection will then be performed through **Significance threshold (%)**.

2.4.2 Instructions and R Code

2.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

2.4.2.2 R code

Role: displays the R script corresponding to the analysis in an **R Code** output. **Impact on the computation:** none. **Practical tip:** enable this option if you want to reproduce the analysis in the jamovi **Rj Editor** or continue working in R. It is **disabled by default**.

2.4.3 Significance threshold: Significance threshold (%)

Role: sets the *p-value* threshold used to select discriminating attributes and the attributes characteristic of each stimulus. **Impact on the computation:** this threshold controls both which attributes appear in **Identification of the Sensory Dimensions** (global test) and which attributes appear in **Sensory Description of the Stimuli** (stimulus-level tests). A low threshold produces more conservative results; a higher threshold retains more attributes in both tables. **Practical tip:** the default value is **5%**, a good starting point for a first trial. If the results are too sparse, try **10%**, then possibly **20%** for exploratory work.

2.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, that the subject variable is categorical, and that the sensory attributes are numeric.
3. Open **SEDA > Fixed List of Attributes > Characterization of the Stimulus Space**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the subject variable in **Subject Effect**.
6. Place all sensory attributes in **Sensory Attributes**.
7. Leave **Significance threshold (%)** at **5%** for a first trial.
8. First read **Identification of the Sensory Dimensions** to identify the attributes that structure the sensory space.
9. Then read **Sensory Description of the Stimuli** to understand the characteristics specific to each stimulus.
10. If no results appear, or if the second table is very sparse, test a slightly broader threshold.

2.6 Interpretation of results

2.6.1 Identification of the Sensory Dimensions

This first table lists the sensory attributes that **significantly discriminate** the stimuli, sorted from the most discriminating to the least discriminating. It is the entry point for interpretation: it indicates which attributes structure the sensory space as a whole.

Element	Meaning
<i>(first column)</i>	Discriminating sensory attribute
Vtest	Standardised measure of the discriminating effect — the higher the value, the stronger the signal
p	P-value of the global test — the lower it is, the more clearly the attribute differentiates the stimuli

The first rows of the table correspond to the most structuring attributes.

Point of vigilance: an attribute absent from this table is not sufficiently discriminating at the chosen threshold. If no attribute is retained, the function displays an error message and does not produce interpretable results.

2.6.2 Sensory Description of the Stimuli

This second table describes each stimulus through the attributes that significantly characterise it. Rows are grouped by stimulus, and the most salient attributes appear first within each group.

Element	Meaning
<i>(first column)</i>	Name of the stimulus
Descriptor	Characteristic sensory attribute
Coefficient	Estimated deviation from the global reference — positive if the stimulus is perceived more intensely, negative otherwise
Adjusted mean	Adjusted mean of the stimulus on this attribute, corrected for the subject effect
p	P-value of the characterisation test
Vtest	Standardised intensity of the association — positive = over-represented, negative = under-represented

- A positive **V-test** suggests that the stimulus is perceived as more intense than the reference, that is, the mean, on this attribute.
- A negative **V-test** suggests the opposite — and is just as informative.
- Use **Adjusted mean** to read the sensory level, but never interpret it without **p** and **Vtest**.

Point of vigilance: a stimulus absent from the table does not have “no profile”; it is simply not significantly different from the mean of the full set of stimuli at the chosen threshold.

In the current configuration of the function, no graphical output is produced. Interpretation is based entirely on the two result tables.

2.7 Example: the sensochoc dataset

The **sensochoc** dataset describes 6 chocolates evaluated by 29 subjects on 14 sensory attributes. It is used to illustrate the function outputs. The numerical values discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

2.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Sensory Attributes	all numeric sensory attributes
Significance threshold (%)	5

2.7.2 Analysis of the *Identification of the Sensory Dimensions* table

With these settings, all 14 attributes emerge as discriminating at the 5% threshold. The table is sorted by decreasing V-test: the attributes at the top are those that most clearly differentiate the chocolates from one another. *MilkF* comes first (V-test = 16.40), followed by *CocoaF* (13.48) and *Bitterness* (13.34). At the bottom of the table, *Sticky* (V-test = 3.56) and *Granular* (4.37) are significant but less structuring. This ranking provides a first reading of the sensory space: the contrasts between chocolates are based above all on milky, cocoa, and bitter dimensions.

2.7.3 Analysis of the *Sensory Description of the Stimuli* table

The second table makes it possible to read the profile of each chocolate relative to the overall mean of all six chocolates.

choc1 is characterised by a highly contrasted profile. For this chocolate, on the positive side, *Bitterness* has the highest V-test (9.95, coefficient +2.46) and an adjusted mean of 7.07, clearly above the overall mean. *CocoaF* (V-test = 8.48), *Astringency* (7.08), and *Acidity* (6.14) complete this profile. On the negative side, *Sweetness* (V-test = -8.29, coefficient -1.95, adjusted mean 3.14) and *MilkF* (V-test = -8.70) indicate a chocolate that is very low in sweetness and very low in milky notes.

choc3 presents the opposite profile. It is strongly characterised by *MilkF* (V-test = 16.52, the highest in the table), *Caramel* (11.52), *Sweetness* (10.39), and *Melting* (9.19). Conversely, it is negatively characterised by *Bitterness* (-12.42), *Crunchy* (-12.49), and *CocoaF* (-13.33). It is the milkiest, sweetest, and most melting chocolate, and the least bitter and cocoa-like. Given the number of significant sensory attributes, **choc3** appears to be the chocolate whose profile differs most clearly from the mean.

choc2 stands out mainly through *Crunchy* (V-test = 6.92) and a lower perception of *MilkF* (-5.16). **choc6** is also *Crunchy* (5.35) and slightly more *Sweetness* (2.41), but less *Melting* (-3.08).

choc4 shows a moderate and weakly contrasted profile — except for *MilkF*, no V-test exceeds 4 in absolute value — which suggests a chocolate close to the general sensory mean on most attributes.

choc5 has very few salient characteristics: only *CocoaA*, *Crunchy*, and *CocoaF* appear slightly on the positive side, and *Sticky* on the negative side. Its profile is the least distinctive of the six chocolates at the 5% threshold.

2.7.4 Synthetic interpretation

This example shows that the six chocolates cover a broad sensory space, with a very bitter and intense chocolate at one extreme (**choc1**) and a milky, sweet, and melting chocolate at the other (**choc3**). Between these two poles, the other chocolates occupy more intermediate positions. The first table identifies the attributes that globally structure the sensory space, while the second specifies the profile specific to each stimulus. The joint reading of these two outputs is the main contribution of the function.

What we take away from this example: the analysis combines a global reading of discriminating attributes and a local reading of stimulus profiles, making it possible to characterise the sensory space in detail.

2.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured with **System R** as described in Chapter 1, check **R code** to display the script corresponding to the analysis. The dataset currently open in jamovi is available as **data**.

For this analysis, the script loads **SensoMineR**, prepares the selected variables, and then calls `decat()` with a model including the stimulus and subject effects. The value of **Significance threshold (%)** is passed to the `proba` argument: 5% in the interface becomes 0.05 in R. The objects `res_QDA$resF` and `res_QDA$resT` correspond respectively to the two tables discussed in this chapter.

This output provides a way to inspect and reproduce the computation and, when useful, continue the analysis in R. It is not required to use SEDA.

2.9 Practical tips

- Start with **Significance threshold (%) = 5%**.
- If no attribute is retained, test **10%**, then **20%** for exploratory work.
- Always read the global table before the stimulus-by-stimulus table.

- Do not confuse a globally discriminating attribute with an attribute that strongly characterises **all** stimuli: an attribute can differentiate stimuli from one another without all of them being strongly associated with it.
- A negative **V-test** is informative: it indicates that a stimulus is perceived as less intense than the mean on an attribute, which can be a sensory characteristic in its own right.
- Use clear and consistent stimulus names to make the results easier to read.
- If your panel includes repetitions, keep all rows in the file.
- Do not overinterpret the absence of a stimulus from the second table: it may simply be located close to the general sensory mean.

2.10 In summary

This analysis first identifies the sensory attributes that globally discriminate the stimuli, then describes each stimulus through the attributes that distinguish it from the global reference, that is, the mean. It takes into account both the stimulus effect and the subject effect, which makes it more informative than a simple comparison of raw means. Its main value is that it combines a **global** reading of the sensory space and a **local** stimulus-by-stimulus reading, in two directly interpretable tables.

3 QDA Data — Multivariate Representation with Ellipses

3.1 Introduction

This analysis is designed for **Quantitative Descriptive Analysis (QDA)** data, that is, when several subjects evaluate several stimuli using a fixed list of sensory attributes. It extends the univariate characterisation of the sensory space by providing a multivariate representation of the stimuli and their variability. It answers two complementary questions: **which stimuli are sensorially similar** and **is this proximity stable despite variability between subjects?** Its central output is a map of the stimuli enriched with **ellipses**, which visualise not only the position of the stimuli, but also the robustness of this position under panel resampling.

Why not use a standard alternative?

A PCA applied directly to the raw data may mix two sources of variation: the **stimulus** effect and the **subject** effect. In sensory analysis, some subjects systematically rate higher or lower than others, or use the scale more or less broadly. This analysis takes this subject/stimulus structure into account to produce a representation better suited to QDA data.

3.2 Understanding what happens under the hood

3.2.1 Selection of sensory attributes

Before constructing the map, the function identifies the attributes that discriminate the stimuli sufficiently. This selection depends on the value of **Descriptor selection threshold (%)**. The stricter the threshold, the more streamlined the map will be; the broader it is, the more exhaustive the representation will be, although sometimes more crowded. The default value is 20%, which makes it a good starting point for exploration.

3.2.2 Correction of the subject effect

A simple way to understand the next step is to say that the analysis reconstructs **mean sensory profiles by stimulus**, while taking differences between subjects into account. Two options are involved here:

- **Center by subject** corrects differences in average level between subjects;
- **Scale by subject** corrects differences in the amplitude with which subjects use the scale.

The objective is to prevent a very generous, very severe, or very “spread-out” subject from artificially distorting the map of the stimuli.

3.2.3 Principal component analysis

A **PCA** is then performed on the resulting table of sensory profiles. It produces the stimulus map, the sensory attribute map, the eigenvalue table, and the automatic description of the dimensions. This factor space is then used as the reference space for projecting the virtual panels.

3.2.4 Virtual panels and supplementary individuals

The key point of the function is the use of **resampling**. In PCA, it is possible to distinguish between **active individuals** — those used to construct the axes — and **supplementary individuals** — those projected afterwards, without modifying the axes. This is precisely the mechanism that makes it possible to construct the ellipses.

At each simulation, a subset of subjects is drawn with replacement from the original panel, new mean stimulus profiles are recalculated, and then projected into the factor space as supplementary individuals. The dispersion of these projections around each stimulus is then summarised by an ellipse.

These ellipses therefore describe the variability of the **mean position of the stimuli when the composition of the panel changes**, not the simple raw dispersion of individual ratings.

Note: the computation may take a few seconds, because it relies on a large number of virtual panels. During this computation, it is better to avoid changing other options so as not to restart the procedure unnecessarily.

3.2.5 Comparison between stimuli

The function also computes a comparison between pairs of stimuli, presented as a matrix of **p-values**. The Hotelling test is performed on the **selected factor plane** defined by **X-axis** and **Y-axis**. This output complements the visual reading of the ellipses; the pairwise comparisons are exploratory and are reported without multiplicity adjustment.

3.2.6 Automatic description of the dimensions

Finally, the function provides an automatic description of the axes, filtered according to the value of **Significance threshold (%)**. It helps identify which attributes are most strongly associated with each dimension and gives a *sensory* meaning to the axes.

3.3 Expected data structure

The data must contain:

- one column identifying the **stimuli**;
- one column identifying the **subjects**;
- at least two numeric columns corresponding to the **sensory attributes**.

Each row corresponds to the evaluation of one stimulus by one subject. If a subject evaluates the same stimulus several times (repetitions or sessions), each evaluation must appear on a separate row.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variable intended for **Subject Effect** is categorical;
- all variables intended for **Sensory Attributes** are numeric.

3.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**. The interface includes three variable drop zones, an **Instructions and R Code** section containing the **Read me before running** and **R code** checkboxes, and then three collapsible menus: **Graphic Options**, **Resampling Options**, and **Numerical Indicators**.

3.4.1 Variables to complete

3.4.1.1 Stimulus Effect

Role: identifies the stimuli to be represented on the map. **Impact on the computation:** each level of this variable becomes a mean point and, where applicable, an ellipse on the map. **Practical tip:** use clear labels, because they will appear directly in the graphics.

3.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is essential for correcting the subject effect and for generating the virtual panels used in the bootstrap. **Practical tip:** check that the same subject keeps exactly the same identifier throughout the file.

3.4.1.3 Sensory Attributes

Role: groups together the sensory attributes used to construct the factor space. **Impact on the computation:** only numeric attributes are accepted; the number of calculable dimensions depends on the number of retained attributes. **Practical tip:** start by including all relevant sensory attributes, then refine the selection with **Descriptor selection threshold (%)**.

3.4.2 Instructions and R Code

3.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

3.4.2.2 R code

Role: displays the R script corresponding to the analysis in an **R Code** output. **Impact on the computation:** none. **Practical tip:** enable this option to reproduce the analysis in the **Rj Editor** or continue working in R. It is **disabled by default**.

3.4.3 Standardization of the Sensory Attributes: Scale to unit variance

Role: standardises the sensory attributes when performing the PCA. **Impact on the computation:** when checked, each attribute contributes equally to the construction of the axes; when unchecked, the most dispersed attributes have a greater influence on the construction of the axes. **Practical tip:** leave this option checked in most analyses. It is **enabled by default**.

3.4.4 Selection of the Sensory Attributes: Descriptor selection threshold (%)

Role: sets the threshold for selecting attributes according to their relationship with the stimulus effect. **Impact on the computation:** a low threshold retains few attributes and produces a more readable map; a higher threshold retains more attributes and gives a broader view. **Practical tip:** the default value is 20%, a good starting point for exploration. For a stricter version, start at 5%.

3.4.5 Graphic Options

3.4.5.1 Components to Plot — X-axis and Y-axis

Role: choose the dimensions displayed on the horizontal and vertical axes. **Impact on the computation:** they do not change the PCA itself, but they modify the displayed factor plane. The selected dimensions must exist in the analysis, otherwise the function generates an error. **Practical tip:** start with X-axis = 1 and Y-axis = 2, the default values.

3.4.5.2 Individual variability around stimuli

Role: adds a graph showing the dispersion of individual evaluations around each stimulus. **Impact on the computation:** it does not add a new analysis, but provides an additional graphical output useful for assessing consensus between subjects. **Practical tip:** activate this option if you want to distinguish a consensual stimulus from a stimulus perceived more variably. The default value is **disabled**.

3.4.5.3 Variability around sensory attributes

Role: adds a graph showing the variability of sensory attributes under resampling. **Impact on the computation:** it provides a graphical output useful for assessing the stability of attribute interpretation. **Practical tip:** activate this option when you want to comment on the robustness of the axes. The default value is **disabled**.

3.4.6 Resampling Options

3.4.6.1 Subjects per virtual panel

Role: defines the number of subjects drawn in each virtual panel. **Impact on the computation:** the closer this value is to the actual panel size, the more similar the simulated panels will be to it. **Practical tip:** use the actual panel size as a starting point. The default value is 20.

3.4.6.2 Number of virtual panels

Role: sets the total number of simulated virtual panels. **Impact on the computation:** the larger this number, the more stable the ellipses will be from one run to another, but computation time increases. **Practical tip:** 300 is a good compromise to start with; for a final rendering, it may be useful to increase this number to 500. The default value is 300.

3.4.6.3 Ellipse alpha (%)

Role: defines the alpha level used to draw the confidence ellipses. **Impact on the computation:** a lower alpha produces wider ellipses; with the default value of 5%, the ellipse corresponds to a 95% level in the covariance-based construction of the simulated positions. **Practical tip:** leave 5% for a first trial.

3.4.6.4 Center by subject

Role: centres responses subject by subject. **Impact on the computation:** corrects differences in average level between subjects and often improves the comparison between profiles. **Practical tip:** leave this option checked in almost all analyses. It is **enabled by default**.

3.4.6.5 Scale by subject

Role: standardises responses subject by subject. **Impact on the computation:** corrects differences in amplitude between subjects, but modifies the initial structure more strongly than centring alone. **Practical tip:** leave this option unchecked at first. It is **disabled by default**.

3.4.7 Numerical Indicators

3.4.7.1 Significance threshold (%)

Role: defines the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the automatic description will be. **Practical tip:** the default value, 5%, is well suited to a first reading.

3.4.7.2 Number of dimensions

Role: defines the number of dimensions included in the automatic description. **Impact on the computation:** SEDA computes enough components to describe these dimensions; the displayed plane is selected separately with **X-axis** and **Y-axis**. **Practical tip:** the default value, 2, is appropriate for an initial reading.

3.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, the subject variable is categorical, and the sensory attributes are numeric.
3. Open **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the subject variable in **Subject Effect**.
6. Place all sensory attributes in **Sensory Attributes**.
7. Leave **Scale to unit variance** checked.
8. Set **Descriptor selection threshold (%)** to 20% for an initial exploration, or to 5% for a stricter selection.
9. Keep **X-axis = 1** and **Y-axis = 2**.
10. Leave **Center by subject** checked and **Scale by subject** unchecked at first.
11. Set **Subjects per virtual panel** to the panel size or a close value, leave **Number of virtual panels** at 300 and **Ellipse alpha (%)** at 5.
12. Activate **Individual variability around stimuli** and/or **Variability around sensory attributes** according to your needs.
13. In **Numerical Indicators**, keep the default values for a first trial.
14. First read **Representation of the Stimuli** and **Representation of the Sensory Attributes**, then **Eigenvalue Decomposition** and **Automatic Description of the Dimensions**. Next assess stability with **Representation of the Stimuli with Confidence Ellipses** and **Hotelling Test**.
15. Then return to the settings to test the stability of your conclusions.

3.6 Interpretation of results

3.6.1 Representation of the Stimuli with Confidence Ellipses

This is the central graph of the function. Each stimulus is represented by a mean point in the factor space, surrounded by an ellipse. The axes come from the PCA, and the percentage displayed on each axis indicates the share of variance explained.

Element	Meaning
Two close stimuli	Similar sensory profiles
Strongly overlapping ellipses	Less clear visual separation under resampling
Small ellipse	Position stable across simulated panels
Large ellipse	More uncertain position
Clearly separated ellipses	Clearer visual separation under resampling

Point of vigilance: the ellipses describe the variability of the mean position of the stimuli under panel resampling. They do not directly represent the raw dispersion of individual ratings. Ellipse overlap is a visual guide, not a significance test; use the **Hotelling Test** for the pairwise statistical comparison.

3.6.2 Representation of the Stimuli

This map shows the relative positions of the stimuli without emphasizing the ellipses. It is useful for quickly identifying the main groupings and oppositions between stimuli.

3.6.3 Representation of the Sensory Attributes

This map shows the sensory attributes in the factor space. It is read using three simple cues:

- **direction:** toward which stimuli the attribute “pulls” the structure;
- **length:** the longer an attribute is, the better it is represented on the plane;
- **the angle between two attributes:** small angle = positive correlation; angle close to 180° = opposition; angle close to 90° = near-independence.

This is the map that gives sensory meaning to the axes.

3.6.4 Individual variability around stimuli (*available only if the option is activated*)

This output shows the dispersion of individual evaluations around the mean point of each stimulus. A compact cloud indicates strong consensus; a dispersed cloud indicates a more heterogeneous perception between subjects.

3.6.5 Variability of the Sensory Attributes (*available only if the option is activated*)

This output shows the dispersion of the position of sensory attributes in the virtual panels. A compact cloud means that the attribute is stable; a dispersed cloud indicates that its interpretation is more fragile and depends more strongly on the composition of the panel.

3.6.6 Eigenvalue Decomposition

This table gives, for each dimension, the eigenvalue, the percentage of variance explained, and the cumulative percentage. If the first two dimensions explain a large share of the variance, the 2D map summarises the data well. Otherwise, it is prudent to explore other planes, for example Dim 1 / Dim 3.

3.6.7 Automatic Description of the Dimensions

This table identifies the attributes most strongly linked to each dimension, filtered by **Significance threshold (%)**. It reports, in particular, their correlation with the dimension and the associated p-value. A dimension requested in **Number of dimensions** may not appear if no attribute reaches the selected significance threshold. The table helps give the axes a sensory interpretation and complements the attribute map.

3.6.8 Hotelling Test

This p-value matrix compares each pair of stimuli on the **selected factor plane** defined by **X-axis** and **Y-axis**. A low p-value provides evidence of separation between two stimuli on that plane; a high p-value does not establish such separation. These pairwise comparisons are exploratory and are reported without multiplicity adjustment. The table therefore complements the visual reading of the ellipses rather than replacing it.

3.7 Example: the sensochoc dataset

The **sensochoc** dataset describes 6 chocolates evaluated by 29 subjects on 14 sensory attributes. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

3.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Sensory Attributes	all sensory attributes
Scale to unit variance	Checked
Descriptor selection threshold (%)	20
X-axis / Y-axis	1 / 2
Subjects per virtual panel	20
Number of virtual panels	300
Ellipse alpha (%)	5
Center by subject	Checked
<i>Scale by subject</i>	Unchecked
Significance threshold (%)	5
Number of dimensions	2

Reproducibility note: the numerical results and resampling-based plots discussed below were generated with the default value **Subjects per virtual panel** = **20**. Use 20 to reproduce this chapter exactly. Because the actual panel contains 29 subjects, **29** is a natural choice for a new final analysis; changing this value changes the resampling-based outputs.

3.7.2 Analysis of the *Representation of the Stimuli with Confidence Ellipses* graph

The map shows the 6 chocolates in the Dim. 1 / Dim. 2 plane. The first dimension alone accounts for 88.79% of the total variance — this 2D plane is therefore a very faithful representation of the data. The second dimension explains an additional 7.58%.

On the horizontal axis (Dim. 1), **choc1** and **choc3** occupy the two opposite extremes: **choc1** on the left (bitter, cocoa side), **choc3** on the right (milky, sweet side). These are the two

chocolates whose profiles are most clearly opposed, which is consistent with the results of the previous chapter. **choc2**, **choc5**, and **choc6** are grouped in the centre, close to the origin. **choc4** is located slightly above the horizontal axis, between **choc1** and the central group.

The ellipses provide information about the robustness of these positions. The ellipses of **choc2**, **choc5**, and **choc6** overlap widely: their visual separation is therefore less clear when panel composition changes. Conversely, **choc1** and **choc3** have ellipses clearly separated from all the others: their visual separation appears robust across the simulated panels. The **Hotelling Test** then provides the statistical assessment of these differences on the displayed plane.

3.7.3 Analysis of the *Hotelling Test* table

The *p-value* table quantitatively complements the ellipse reading on the Dim. 1 / Dim. 2 plane. The pairs **choc2** / **choc5** ($p = 0.76$) and **choc5** / **choc6** ($p = 0.16$) have high *p*-values: no separation is established between these chocolates on this plane. The pair **choc2** / **choc6** is more borderline ($p = 0.04$). Conversely, all comparisons involving **choc1** or **choc3** give *p*-values of 0.00 in the results of this example. These pairwise comparisons are exploratory and are not adjusted for multiplicity.

3.7.4 Analysis of the *Representation of the Sensory Attributes* graph

The 14 attributes are distributed very clearly around the horizontal axis. On the right are grouped *MilkF*, *Sweetness*, *Caramel*, *Vanilla*, *MilkA*, and *Melting* — this is the profile of **choc3**. On the left are grouped *CocoaF*, *CocoaA*, *Bitterness*, *Astringency*, *Acidity*, *Crunchy*, and *Granular* — this is the profile of **choc1**. The horizontal axis therefore clearly corresponds to an opposition between **sweet milk chocolate** / **bitter dark chocolate**.

The length of the arrows also indicates the quality of representation of each attribute on this plane. *Crunchy* has a shorter arrow pointing downward, indicating that it is less well represented on this plane than *CocoaF* or *MilkF*, whose arrows almost reach the edge of the circle. Arrow length does not, by itself, measure stability across panels; that question is addressed by the **Variability of the Sensory Attributes** plot.

3.7.5 Analysis of the *Automatic Description of the Dimensions* table

The automatic description of the dimensions confirms the very strong opposition carried by Dim. 1. The most positively correlated attributes are *MilkF* (1.00), *MilkA* (0.99), *Caramel* (0.99), *Vanilla* (0.98), and *Sweetness* (0.95). At the opposite end, the strongest negative correlations are observed for *CocoaF* (-1.00), *Bitterness* (-0.97), *CocoaA* (-0.97), *Granular* (-0.96), and *Astringency* (-0.94). With **Significance threshold (%) = 5**, no attribute is retained for Dim. 2 in this table.

3.7.6 Synthetic interpretation

The sensory space of the chocolates is very largely structured by a single strong opposition, carried by the first dimension: on one side, a milky, sweet, and melting profile; on the other, a bitter, cocoa, and more textured profile. The stimulus map and the attribute map converge toward the same overall reading. The ellipses additionally show that not all stimuli are differentiated with the same robustness. Finally, the Hotelling test confirms that some proximities observed on the map do not correspond to clearly established differences.

What we take away from this example: the analysis does not only show where the stimuli are located in the sensory space, but also how robustly they can be distinguished.

3.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured to use **System R** as described in Chapter 1, check **R code** to display the script corresponding to the analysis. For this chapter, the code loads **SensoMineR** and **FactoMineR**, prepares the selected variables, and then calls **panellipse()** using the settings chosen in the interface.

The generated script also includes a small helper that prevents **panellipse()** from opening an additional graphics device that is not supported by the Rj Editor. This adaptation is handled automatically: no additional setting is required.

The resulting objects provide access, among other things, to eigenvalues, Hotelling tests, and the automatic description of dimensions. The code therefore provides a reproducible continuation of the analysis performed in SEDA.

3.9 Practical tips

- Always start with X-axis = 1 and Y-axis = 2.
- Leave **Scale to unit variance** checked in most cases.
- Use **Center by subject** almost systematically.
- Activate **Scale by subject** only if you have a good reason to think that subjects use the scale with very different amplitudes.
- Start with 300 simulated panels; increase to 500 or more for a more stable final version.
- Set **Descriptor selection threshold (%)** between 5% and 20% depending on whether you want a streamlined or more exhaustive map.
- Never interpret the stimulus map without looking at the attribute map.
- Do not conclude that there is a difference from ellipse overlap or separation alone.

- Consult **Hotelling Test** to complement the visual reading when the overlap is ambiguous.
- If the first two dimensions summarise little variance, explore other factor planes.

3.10 In summary

This analysis produces a multivariate representation of stimuli suited to QDA data, while taking the subject effect into account in the construction of sensory profiles. It enriches the classical factor map with ellipses obtained by resampling, making it possible to assess the robustness of the observed positions. It therefore shows not only **where the stimuli are located**, but also **how confidently they can be located there**.

4 CATA Data — Analysis of Checked Terms

4.1 Introduction

This analysis is designed for **CATA** (*Check-All-That-Apply*) data, that is, when subjects check, from a predefined list, all the terms they consider appropriate to describe a stimulus. It answers two complementary questions: **which terms characterise each stimulus** and **how the stimuli are globally organised according to their CATA profiles**. The function produces a contingency table, a description of the stimuli according to the CATA terms, as well as a **correspondence analysis** followed by a **classification of the stimuli**, automatic by default.

Why not use a standard alternative?

CATA data are originally binary responses by subject and by term. A classical comparison of means is therefore not the most suitable tool for representing the global structure of these data. The function first aggregates the responses into a **stimuli $\times$ terms contingency table**, then applies methods suited to this structure: frequency description, correspondence analysis, and, when possible, classification.

4.2 Understanding what happens under the hood

4.2.1 Construction of the contingency table

The first step consists of grouping the responses by stimulus. The function sums the columns placed in the **CATA** field for each level of **Stimulus Effect**. This produces a table where rows are stimuli, columns are CATA terms, and each cell indicates how many times a term was checked for a given stimulus. This table is the basis for all subsequent outputs.

Note: this table can be copied into a spreadsheet file.

4.2.2 Description of stimuli by terms

From this contingency table, the function applies a frequency description procedure based on the threshold value defined in **Significance threshold (%)**. For each stimulus, it identifies the terms that are significantly more or less frequent than in the whole table. This logic is close to a statistical description of profiles, but adapted here to *checking* frequencies rather than to continuous ratings. The same threshold is then applied to the description of clusters.

A term absent from the description table is not significantly associated with a stimulus at the chosen threshold. This may mean that it is checked fairly evenly across stimuli.

4.2.3 Correspondence analysis

If at least **three** variables are placed in the **CATA** field, the function performs a **correspondence analysis** on the contingency table. This method simultaneously represents the stimuli and the terms in the same factor space. Two stimuli that are close on the map tend to share similar CATA profiles; two terms that are close tend to be checked for similar stimuli. The axes summarise the main profile oppositions between stimuli and terms.

4.2.4 Classification of stimuli

Once the correspondence analysis has been computed, the function runs a **hierarchical clustering on principal components (HCPC)**. By default, **Number of clusters = -1** lets FactoMineR determine the number of groups automatically, but the user can also impose a number of clusters. The clustering uses the first CA dimensions defined by **Number of dimensions for clustering**; this setting is distinct from the factor plane displayed with **X-axis** and **Y-axis**. The percentages displayed on the HCPC map are expressed relative to the inertia of this retained clustering space and may therefore differ from those on the full CA map.

The function then produces a map of the stimuli according to their clusters and a description of the clusters using the same significance threshold as for the description of stimuli.

4.2.5 Minimum conditions for running the analysis

SEDA requires at least **three CATA variables**. With fewer than three attributes, the analysis stops. To construct a two-dimensional correspondence-analysis map, the aggregated table must also contain at least **three stimuli** and **three informative CATA attributes**.

CATA variables must be complete **0/1** indicators: 0 means that the term was not checked and 1 that it was checked. Missing values are not interpreted as unchecked responses and must be resolved before the analysis.

4.3 Expected data structure

The data must contain:

- one categorical column identifying the **stimuli**;
- at least **three CATA columns** coded as 0/1.

Each row corresponds to the evaluation of one stimulus by one subject. If 40 subjects evaluate 6 stimuli, the file generally contains one row per subject and per stimulus, with one column per *CATA* term.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is categorical;
- the variables intended for **CATA** are numeric and contain only 0 and 1;
- no CATA response is missing: a missing value is not treated as 0;
- at least **three** CATA terms are selected;
- at least **three stimuli** are actually observed to obtain the two-dimensional factor map.

4.4 Interface guide

The analysis appears in the menu **SEDA > Fixed List of Attributes > Analysis of CATA Data**. The interface includes two variable drop zones, an **Instructions and R Code** section, the **Significance threshold (%)** field, and two collapsible menus: **Graphic Options** and **Clustering**.

4.4.1 Variables to complete

4.4.1.1 Stimulus Effect

Role: identifies the stimuli to compare. **Impact on the computation:** this variable defines the rows of the contingency table and organises all descriptive and graphical outputs. **Practical tip:** use clear labels, because they will appear in the tables and graphs.

4.4.1.2 CATA

Role: groups together the terms from the CATA list. **Impact on the computation:** these variables define the columns of the contingency table; they are then used for stimulus description, correspondence analysis, and clustering. **Practical tip:** select at least **three** complete CATA variables coded as 0/1.

4.4.2 Instructions and R Code

4.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

4.4.2.2 R code

Role: displays the R script corresponding to the analysis in an **R Code** output. **Impact on the computation:** none. **Practical tip:** enable this option to reproduce the analysis in the **Rj Editor** or continue working in R. It is **disabled by default**.

4.4.3 Significance threshold (%)

Role: defines the p-value threshold used to select terms characterising stimuli and clusters. **Impact on the computation:** this threshold affects the **descfreq** description tables but does not change the construction of the correspondence analysis or the clustering. **Practical tip:** the default value is 5%.

4.4.4 Graphic Options

4.4.4.1 X-axis and Y-axis

Role: choose the two correspondence-analysis dimensions displayed on the maps. **Impact on the computation:** they change the displayed plane, not the CA structure itself. **Practical tip:** start with X-axis = 1 and Y-axis = 2, the default values.

4.4.5 Clustering

4.4.5.1 Number of dimensions for clustering

Role: sets the number of first CA dimensions used by HCPC. **Impact on the computation:** clustering may therefore use more information than the displayed plane alone. The default value is 5; if fewer than five dimensions are available, SEDA uses all available dimensions. **Practical tip:** the axes chosen for the graph must be included among the dimensions used for clustering.

4.4.5.2 Number of clusters

Role: sets the number of stimulus groups. **Impact on the computation:** with the default value -1, FactoMineR determines the number of clusters automatically; an integer value of at least 2 imposes that number of groups. The requested number must remain lower than the number of stimuli. **Practical tip:** start with -1, and impose a number of groups only when there is an analytical reason to do so.

4.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical and that the CATA variables are complete and coded 0/1.
3. Open **SEDA > Fixed List of Attributes > Analysis of CATA Data**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place at least three terms in **CATA**.
6. Leave **Significance threshold (%)** at 5% for a first trial.
7. Keep X-axis = 1 and Y-axis = 2.
8. Leave **Number of dimensions for clustering = 5** and **Number of clusters = -1** for an initial analysis.
9. Start by reading **Description of the Stimuli According to CATA**.
10. Then consult **Representation of the Stimuli and the CATA**, interpreting stimuli and terms relative to the axes.
11. Read **Representation of the Stimuli According to Clusters** and then **Description of the Clusters**.
12. Finally check **Aggregated CATA Table — Stimuli $\times$ CATA Frequency Table**, which contains the frequencies actually used in the computations.
13. If necessary, change the factor plane or clustering settings, then check that your conclusions remain coherent.

4.6 Interpretation of results

4.6.1 Contingency table

This first table crosses stimuli in rows and CATA terms in columns. Each cell indicates the number of times a term was checked for a given stimulus.

Element	Meaning
High value in a cell	The term was often checked for this stimulus

Element	Meaning
Row with several high values	The stimulus is associated with several frequent terms
Column high for almost all stimuli	Frequent term, but not necessarily discriminating

Point of vigilance: a very frequent term is not automatically a characteristic term. If it is checked for almost all stimuli, it will contribute little to distinguishing them.

4.6.2 Description of the Stimuli According to CATA

This table describes each stimulus through the terms that are significantly associated with it, grouped by stimulus. Terms are sorted by decreasing V-test within each group.

Element	Meaning
<i>(first column)</i>	Stimulus concerned
CATA	Characteristic term
Intern %	Percentage of use of the term for this stimulus
Global %	Overall percentage of use of the term
Intern frequency	Absolute frequency of the term for this stimulus
Global frequency	Total absolute frequency of the term
p	P-value of the test
V-test	Standardised intensity of the association

- A positive **V-test** indicates that the term is checked more frequently for this stimulus than overall.
- A negative **V-test** indicates that it is checked less frequently — and is just as informative.
- A low **p-value** indicates a clearer characterisation.

A stimulus absent from this table does not have “no profile”; it is simply close to the mean profile of all stimuli at the chosen threshold.

4.6.3 Representation of the Stimuli and the CATA (*available with at least 3 terms*)

This graph is produced only if at least **three** terms are placed in **CATA**. It simultaneously represents the stimuli and the terms in the factor space derived from the correspondence analysis.

- Two close stimuli have similar CATA profiles.

- Stimuli and terms should be interpreted relative to the **factor dimensions** and the oppositions they structure; a direct distance between a stimulus and a term should not be read as an ordinary Euclidean distance.
- The percentage displayed on each axis indicates the share of total information captured by that dimension.

This map gives an overview, but it does not replace the descriptive table. The two should be read together.

4.6.4 Representation of the Stimuli According to Clusters (*available with at least 3 terms*)

This graph represents the stimuli according to the HCPC partition. With **Number of clusters = -1**, the partition is selected automatically; if a number of clusters is imposed, the map reflects that partition. Clustering uses the first dimensions defined by **Number of dimensions for clustering**, while **X-axis** and **Y-axis** determine the displayed plane. The percentages on the HCPC map refer to the retained clustering space, not to the full CA inertia.

- Stimuli in the same cluster have similar CATA profiles.
- Distant clusters correspond to more contrasted families of stimuli.
- This output helps summarise the global structure of the stimuli.

4.6.5 Description of the Clusters (*available with at least 3 terms*)

When the classification is available, the function also produces a table describing each cluster through the terms that are significantly associated with it, using the same logic as for the description of stimuli.

Element	Meaning
<i>(first column)</i>	Group of stimuli concerned
CATA	Characteristic term
Intern %	Percentage of the term within the cluster
Global %	Overall percentage of the term
Intern frequency	Absolute frequency of the term within the cluster
Global frequency	Total absolute frequency of the term
p	P-value
V-test	Standardised intensity of the association

This table is particularly useful for summarising the main families of stimuli, beyond the stimulus-by-stimulus details.

4.7 Example: the *Giacalone_et_al_cata* dataset

The **Giacalone_et_al_cata** dataset describes 9 beers evaluated by 74 consumers on 15 CATA items: here, the aim is to associate beers with particular occasions. It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

4.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimulus Effect	Product
CATA	all CATA items from ThirstQ to Concert
Significance threshold (%)	5
X-axis / Y-axis	1 / 2
Number of dimensions for clustering	5
Number of clusters	-1 (automatic)

4.7.2 Analysis of the *Description of the Stimuli According to CATA* table

The table allows a stimulus-by-stimulus reading with test values and actual frequencies.

GMA is significantly characterised by *Rugby* (V-test = 3.95, $p < .001$; 11.61% versus 7.08% overall) and *CampingFishing* (2.73, $p = 0.006$). Conversely, it is under-represented by *Gift* (V-test = -4.36), *Treat* (-3.84), and *FineDinning* (-3.51) — all highly significant.

LR has the profile most strongly marked by *Rugby* in the whole set (V-test = 5.74, $p < .001$; 15.24% versus 7.08% overall) and *CampingFishing* (2.83). It is under-represented by *Guests* (-4.77), *FineDinning* (-3.91), *Treat* (-3.74), and *Gift* (-3.50).

SSA stands out from all other stimuli by the intensity of its positive associations: *Gift* (V-test = 6.43, $p < .001$; 13.41% versus 4.21% overall), *FineDinning* (6.09), *Treat* (4.53), *Achievement* (3.71), and *Guests* (2.72). Conversely, it is strongly under-represented by *Rugby* (-4.29), *ThirstQ* (-3.63), *Work* (-3.01), *CampingFishing* (-2.88), and *TV* (-2.80). It is the stimulus whose profile differs most clearly from the mean.

SC is more strongly linked to *Rugby* (2.82) and *ThirstQ* (2.69), and is under-represented by *PubHouse* (-2.67), *Treat* (-2.58), and *Gift* (-2.55).

HPW is more strongly associated with *Treat* (V-test = 2.41) and *Gift* (2.21), and is under-represented by *Rugby* (-3.29). **PKB** is more strongly associated with *Gift* (2.60) and *Treat* (2.10); no negative term reaches the 5% threshold for this stimulus.

MBB mainly stands out through *PubHouse* (2.62) and is negatively distinguished by *ThirstQ* (-2.35) and *Rugby* (-2.19).

STG is characterised only by *Rugby* on the negative side (-2.75, $p = 0.006$).

4.7.3 Analysis of the *Representation of the Stimuli and the CATA graph*

The first dimension explains 85.59% of the inertia — a remarkably high value, meaning that the main opposition between stimuli is very strong and very readable on the horizontal axis. The second dimension explains only 7.43%, but refines secondary distinctions.

On the map, *SSA* is located very clearly on the right, isolated near *Gift*, *FineDinning*, and *Achievement*. On the left are grouped *LR*, *GMA*, and *SC*, close to *Rugby* and *CampingFishing*. *MBB*, *HPW*, *PKB*, *MHR*, and *STG* occupy a central or slightly right-hand position, between the two poles. On the second dimension, *LR* is located slightly higher than *GMA* and *SC*, which reflects its stronger association with *Rugby*.

4.7.4 Analysis of the *Representation of the Stimuli According to Clusters and the Description of the Clusters graph and table*

The automatic classification retains **three clusters**:

- **Cluster 1** (black): *LR*, *GMA*, and *SC* — positively characterised by *Rugby* (V-test = 9.12, $p < .001$), *CampingFishing* (4.86), *ThirstQ* (3.81), and *TV* (3.01), and under-represented notably by *Gift* (-6.97), *Treat* (-6.81), *FineDinning* (-6.21), and *Guests* (-4.67). This cluster corresponds to the pole most strongly associated with sports and outdoor contexts.
- **Cluster 2** (red): *MBB*, *HPW*, *PKB*, *MHR*, and *STG* — positively characterised by *Treat* (V-test = 3.52), *Guests* (2.86), *Gift* (2.18), *PubHouse* (2.15), and *Achievement* (2.05), and under-represented by *Rugby* (-7.07), *CampingFishing* (-3.34), and *ThirstQ* (-2.02). This cluster groups beverages more associated with social or treat-related contexts and less with sports and outdoor contexts.
- **Cluster 3** (green): *SSA* alone — positively characterised by *Gift* (V-test = 6.43), *FineDinning* (6.09), *Treat* (4.53), *Achievement* (3.71), and *Guests* (2.72), and negatively by *Rugby* (-4.29), *ThirstQ* (-3.63), *Work* (-3.01), *CampingFishing* (-2.88), *TV* (-2.80), and *Party* (-2.39). The profile of *SSA* is singular enough to form a cluster on its own.

4.7.5 Synthetic interpretation

This example clearly shows the value of the function. It does not merely show that an item is often checked: it shows **for which stimuli it is truly characteristic, how the stimuli are globally opposed**, and **which main profiles** emerge from the set of responses. Here, the major opposition — captured at 85.59% by the first dimension — contrasts beverages associated with sports and outdoor contexts (*Rugby*, *CampingFishing*) with beverages perceived as more sophisticated (*Gift*, *FineDinning*, *Treat*). *SSA* is a special case, with a profile singular enough to form a cluster on its own.

What we take away from this example: CATA analysis makes it possible to move from simple checking frequencies to a structured reading of the oppositions between stimuli and the main profiles that emerge from them.

4.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured to use **System R** as described in Chapter 1, check **R code** to display the script corresponding to the analysis. The dataset currently open in jamovi is available as **data**.

For CATA, the script loads **FactoMineR**, aggregates the 0/1 indicators by stimulus, then uses **descfreq()** for characterisation, **CA()** for correspondence analysis, and **HCPC()** for clustering. The **X-axis / Y-axis**, **Number of dimensions for clustering**, and **Number of clusters** settings are carried into the generated code.

The script also uses graphics options suited to the Rj Editor so that no external graphics device is opened. No additional setting is required.

4.9 Practical tips

- Start with **Significance threshold (%) = 5%**; increase to 10% if the description tables are too empty.
- Always check the contingency table before interpreting the graphs.
- At least **three CATA terms and three observed stimuli** are required to construct the two-dimensional factor map.
- A negative **V-test** is just as informative as a positive **V-test**: it indicates that a term is checked significantly less often for a stimulus than for the others.
- Do not overinterpret the absence of a stimulus from the descriptive table: it may simply have a profile close to the mean profile.

- Classification is **automatic by default** (`Number of clusters = -1`), but you can impose a number of groups; always assess its relevance by cross-reading the map and the cluster description.
- Read the correspondence map together with the description table: the map gives an overview, while the table provides the details.
- CATA columns must contain only 0 and 1 and must not contain missing values; a missing value is not treated as an unchecked response.

4.10 In summary

This analysis transforms CATA responses into a structured reading of the CATA profiles of stimuli. It combines a **contingency table**, a **statistical description of stimuli**, a **correspondence analysis**, and a **classification**, automatic by default. Its main value is to move from a simple accumulation of checkmarks to a view of the stimuli that is simultaneously **descriptive**, **factorial**, and **typological**.

5 JAR Data — Defect and Penalty Analysis

5.1 Introduction

This analysis is designed for **JAR** (*Just About Right*) data, in which subjects evaluate a sensory attribute relative to an ideal level, for example *too little* / *just about right* / *too much*. It identifies deviations from the JAR level, estimates their frequency, and studies their relationship with liking.

SEDA provides **two complementary penalty views**. The first is a **global** view, based on **SensoMineR::JAR**, which estimates the importance of defects across the whole product space. The second is a **classical product-specific analysis**, which compares, for a given product, the liking of consumers who rate an attribute JAR with that of consumers who rate it below or above JAR.

Why not use a simple mean liking score?

A mean liking score by product does not show which sensory deviations are associated with a decrease in liking. Likewise, the frequency of JAR responses alone does not indicate whether a frequent defect is actually penalising. The value of the analysis is therefore to combine three different pieces of information: **defect frequency**, **size of the liking loss**, and **statistical uncertainty around that loss**.

5.2 Understanding what happens under the hood

5.2.1 Checking and grouping the JAR coding

The value entered in **Coding of the JAR level** must exactly match the category representing the ideal level in every selected JAR attribute. The default value is **jar**.

For the classical product-specific analysis, SEDA then groups the other categories into two directions:

- **Below JAR;**
- **Above JAR.**

This grouping uses the expressions defined in **Below-JAR keywords** and **Above-JAR keywords**. The first output to inspect is therefore **JAR Coding Used for the Classical Penalty Analysis**. A level shown as **Unclassified** should be resolved before interpreting the classical analysis; otherwise, the corresponding attribute is excluded from that part of the analysis.

5.2.2 Distribution of JAR responses

SEDA calculates, for each product and attribute, the percentages of consumers **Below JAR**, **JAR**, and **Above JAR**. This output describes how frequent the deviations are, but does not yet indicate whether they reduce liking.

5.2.3 Global multidimensional penalties

SensoMineR::JAR estimates penalties using all products simultaneously. In SEDA, they are presented as **Mean drop** values: a positive value means that the defect is associated with a loss of liking across the global product space.

These coefficients are **global** and are not product-specific estimates. SEDA then combines them with the frequency of each defect in each product in **Global Defects Applied to Each Product**.

A row receives **Priority = Yes** when three conditions are met:

- the percentage of consumers reaches **Consumer threshold (%)**;
- the **Global mean drop** reaches **Mean-drop threshold**;
- the global penalty p-value is below 0.05.

The two interface thresholds are therefore **practical decision aids**, not substitutes for the p-value.

5.2.4 Classical product-specific penalty analysis

SEDA also calculates, separately for each product, attribute, and direction (**Below JAR** or **Above JAR**):

Mean drop = Mean liking JAR — Mean liking non-JAR

A positive value therefore means that consumers perceiving the deviation give a lower mean liking score than consumers rating the attribute JAR.

For each comparison, SEDA reports:

- the percentage of consumers concerned;

- mean liking in the JAR group;
- mean liking in the non-JAR group;
- **Mean drop**;
- its 95% confidence interval;
- the p-value.

This analysis is genuinely **product-specific** and complements the global penalty view.

5.2.5 Multivariate synthesis of defects

Finally, SEDA constructs a **products $\times$ non-JAR categories** frequency table, describes it with `descfreq()`, and performs a **correspondence analysis**.

The product description deliberately uses a permissive exploratory threshold (`proba = 0.5`). It is intended to reveal the categories structuring the space and **should not be interpreted as a conventional 5% significance filter**.

The correspondence map summarises the product $\times$ defect structure. As in the CATA chapter, products and defects should be interpreted relative to the factor dimensions; a direct row-to-column distance is not an ordinary Euclidean distance. Proximity on this map is not, by itself, evidence of a liking penalty.

5.3 Expected data structure

The data must contain:

- one categorical column for **Stimulus Effect**;
- one categorical column for **Subject Effect**;
- one numeric column for **Liking Variable**;
- one or more **nominal** columns for **Sensory Attributes**.

Each row corresponds to one subject evaluating one stimulus.

Before running the analysis, check that:

- **Stimulus Effect** contains at least two observed products;
- **Subject Effect** contains at least two observed subjects;
- **Liking Variable** is numeric and contains enough finite observations;
- JAR variables are defined as **Nominal** in jamovi;
- every JAR attribute contains the category specified in **Coding of the JAR level**;
- every JAR attribute has at least two observed response levels;
- the variables used for Stimulus, Subject, Liking, and JAR attributes are distinct.

5.4 Interface guide

The analysis appears in **SEDA > Fixed List of Attributes > Analysis of JAR Data**. The interface includes four variable drop zones, an **Instructions and R Code** section, the **Coding of the JAR level** field, and a **Classical Penalty Analysis** section.

5.4.1 Variables to complete

5.4.1.1 Stimulus Effect

Role: identifies the products to analyse. **Impact on the computation:** this variable defines the products used in both global and product-specific analyses. **Practical tip:** use short, explicit labels.

5.4.1.2 Subject Effect

Role: identifies the subjects. **Impact on the computation:** this variable is used by the global JAR analysis and structures individual observations. **Practical tip:** ensure that each subject keeps the same identifier throughout the dataset.

5.4.1.3 Liking Variable

Role: contains the liking score used to calculate penalties. **Impact on the computation:** all liking losses are expressed in the units of this variable. **Practical tip:** use the same liking scale for all products.

5.4.1.4 Sensory Attributes

Role: groups the JAR variables. **Impact on the computation:** each variable is used in the global, product-specific, and multivariate analyses. **Practical tip:** variables must be **nominal**, even when their categories conceptually follow an order such as *too little / JAR / too much*.

5.4.2 Instructions and R Code

5.4.2.1 Read me before running

Role: displays integrated help. **Impact on the computation:** none. It is **enabled by default**.

5.4.2.2 R code

Role: displays the R script reproducing the analysis. **Impact on the computation:** none. Enable it to continue the analysis in the **Rj Editor**. It is **disabled by default**.

5.4.3 Coding of the JAR level

Role: gives the exact label of the JAR category. **Impact on the computation:** this category is used as the reference in all analyses. **Practical tip:** the default is `jar`; match the label in the data exactly.

5.4.4 Classical Penalty Analysis

5.4.4.1 Consumer threshold (%)

Role: defines the percentage of consumers above which a defect is considered frequent enough for an operational reading. **Default:** 20%. **Important:** this is a decision aid, not a significance threshold.

5.4.4.2 Mean-drop threshold

Role: defines the liking loss above which an effect is considered large enough for an operational reading. **Default:** 1. **Important:** this threshold concerns effect size, not statistical uncertainty.

5.4.4.3 Below-JAR keywords

Role: defines the words or expressions used to recognise categories below the JAR level. The default is `too low|not enough|too weak|plain|low`.

5.4.4.4 Above-JAR keywords

Role: defines the words or expressions used to recognise categories above the JAR level. The default is `too high|too much|too strong|vivid|high`.

Practical tip: always start by checking **JAR Coding Used for the Classical Penalty Analysis**. If a category is **Unclassified**, adapt the keywords before interpreting product-specific results.

5.5 Step-by-step procedure

1. Open the dataset in jamovi.
2. Check that **Stimulus Effect** and **Subject Effect** are categorical, **Liking Variable** is numeric, and JAR attributes are **nominal**.
3. Open **SEDA > Fixed List of Attributes > Analysis of JAR Data**.
4. Fill **Stimulus Effect**, **Subject Effect**, **Liking Variable**, and **Sensory Attributes**.
5. Check **Coding of the JAR level**.
6. Leave **Consumer threshold (%) = 20** and **Mean-drop threshold = 1** for a first reading.
7. Check **Below-JAR keywords** and **Above-JAR keywords**.
8. Read **JAR Coding Used for the Classical Penalty Analysis** and resolve any **Unclassified** category.
9. Read **Distribution of JAR Responses by Product and Attribute**.
10. Examine **Global Multidimensional Penalties (All Products)** and then **Global Defects Applied to Each Product**.
11. Use **Global Penalty Context for Each Product** to see which products are exposed to globally important defects.
12. Then read **Product-Specific Classical Penalty Analysis** and the **Product-Specific Penalty Plots**.
13. Finish with **Description of the Products According to Defects, Representation of the Products and the Defects**, and **Frequency Table Used for the Correspondence Analysis**.
14. Cross-read frequency, mean-drop magnitude, and statistical uncertainty before retaining a product-optimisation lever.

5.6 Interpretation of results

5.6.1 JAR Coding Used for the Classical Penalty Analysis

This table is the essential preliminary check. For each attribute and original response level, it shows whether SEDA uses it as **Below JAR**, **JAR**, **Above JAR**, or **Unclassified**.

An **Unclassified** category should not be ignored: the corresponding attribute is excluded from the classical product-specific analysis until its coding is resolved.

5.6.2 Distribution of JAR Responses by Product and Attribute

For each product and attribute, this table reports:

- **Below JAR (%)**;

- **JAR (%)**;
- **Above JAR (%)**;
- **N**.

It answers the question: **how many consumers actually perceive the deviation?** It does not measure its impact on liking.

5.6.3 Global Multidimensional Penalties (All Products)

This table contains the coefficients from `SensoMineR::JAR`:

Column	Interpretation
Defect	Non-JAR category
Mean drop	Global liking loss associated with the defect
Std. Error	Estimation uncertainty
p	Associated statistical evidence

Mean drop is presented so that a positive value corresponds to a loss of liking.

Point of vigilance: these coefficients are estimated across all products. They should not be described as product-specific penalties.

5.6.4 Global Defects Applied to Each Product

This table combines the local frequency of a defect in each product with its **Global mean drop**.

Column	Interpretation
Consumers (%)	Defect frequency in the product
Global mean drop	Globally estimated mean drop
p	Global penalty p-value
Priority	Yes if frequency, mean drop, and p-value meet the criteria

With the default settings, **Priority = Yes** therefore means at least 20% of consumers concerned, a mean drop of at least 1, and $p < .05$.

5.6.5 Global Penalty Context for Each Product

These graphs provide the native SensoMineR visualisation of the **global** penalty in the context of each product. They help identify products exposed to globally important defects, but the y-coordinate does not thereby become a product-specific penalty.

5.6.6 Product-Specific Classical Penalty Analysis

This table answers a different question: **for this specific product, do consumers perceiving the attribute as non-JAR like the product less than consumers rating it JAR?**

Column	Interpretation
Product	Product concerned
Attribute	JAR attribute
Direction	Below JAR or Above JAR
Consumers (%)	Percentage of consumers in that direction
Mean liking JAR	Mean liking in the JAR group
Mean liking non-JAR	Mean liking in the relevant non-JAR group
Mean drop	Mean liking JAR – Mean liking non-JAR
CI 95% low / high	95% confidence interval for the mean drop
p	Test p-value

A positive **Mean drop** indicates a liking loss associated with the deviation for that product.

5.6.7 Product-Specific Penalty Plots

For each product:

- the x-axis represents **Consumers concerned (%)**;
- the y-axis represents **Mean drop in liking**;
- the vertical dashed line corresponds to the **Consumer threshold**;
- the horizontal dashed line corresponds to the **Mean-drop threshold**;
- symbols distinguish **Below JAR** from **Above JAR**;
- * indicates $p < .05$;
- vertical intervals show the 95% confidence interval.

A defect in the high-frequency / high-mean-drop region may be an operational lever, but confidence intervals and p-values should remain part of the decision.

5.6.8 Description of the Products According to Defects

This table describes the non-JAR categories structuring each product profile. It deliberately uses a very permissive exploratory threshold in this multivariate synthesis. A category displayed with $p > .05$ is therefore not “significant at 5%”; it is retained to help interpret the structure of the space.

5.6.9 Representation of the Products and the Defects

This map comes from correspondence analysis of the products $\times$ defects table. Two nearby products have similar defect profiles. For product-to-defect relationships, interpret directions and factor dimensions rather than a direct row-to-column distance.

This map describes the structure of defects; it is not a test of liking penalty.

5.6.10 Frequency Table Used for the Correspondence Analysis

This table contains the products $\times$ non-JAR-category frequencies used by `descfreq()` and correspondence analysis. It is a useful audit output for understanding exactly which aggregated data feed the multivariate synthesis.

5.7 Example: the milkshake dataset

The **milkshake** dataset contains three milkshakes, **284**, **859**, and **972**, evaluated by 47 consumers using several JAR attributes related to color, fluidity, odors, sweetness, acidity, texture, and taste. To load the dataset, open jamovi, click the icon in the top left corner, then **Open** $\rightarrow$ **Data Library** $\rightarrow$ **SEDA**.

This example is especially useful because it shows that the **global** and **product-specific** views can lead to complementary diagnoses.

5.7.1 Example settings

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Liking Variable	Liking
Sensory Attributes	all JAR variables from Color to Taste_Van
Coding of the JAR level	jar

Option	Value
Consumer threshold (%)	20
Mean-drop threshold	1
Below-JAR keywords	too low not enough too weak plain low
Above-JAR keywords	too high too much too strong vivid high too_fluid

The final term added to **Above-JAR keywords** correctly classifies `Too_fluid` as **Above JAR**. SEDA accepts both `Too_fluid` and `too fluid`: original labels and their normalised forms are recognised.

5.7.2 Check the JAR coding first

The **JAR Coding Used for the Classical Penalty Analysis** table confirms that all response levels used in this example are classified as **Below JAR**, **JAR**, or **Above JAR**. For *Fluidity*, `Not_enough_fluid` is classified as **Below JAR** and `Too_fluid` as **Above JAR**.

This check matters because the classical product-specific analysis can only be interpreted when the non-JAR categories of the attribute are correctly grouped.

5.7.3 Distribution of JAR responses

The **Distribution of JAR Responses by Product and Attribute** table identifies the most frequent deviations.

For **284**, **Below JAR** responses are particularly frequent for *Odor_Van* (59.57%), *Texture* (57.45%), and *Odor_Int* (53.19%), while 46.81% of consumers rate *Fluidity* **Above JAR**.

For **859**, two deviations dominate: 80.85% of consumers are **Below JAR** for *Texture* and 70.21% are **Above JAR** for *Fluidity*.

For **972**, 55.32% are **Below JAR** for *Odor_Int*, 42.55% are **Above JAR** for *Sweetness*, and 38.30% are **Above JAR** for *Fluidity*.

These percentages describe prevalence. They do not by themselves show that the deviations reduce liking.

5.7.4 Global multidimensional penalties

In **Global Multidimensional Penalties (All Products)**, only one defect is significant at the 5% level: *Too_much_odor_fruit*, with **Mean drop = 2.68** and **p = 0.018**.

Several other effects have notable mean drops without reaching the 5% threshold, for example *Too_much_acidity* (4.02, p = 0.082), *Too_much_texture* (1.78, p = 0.089), and *Not_enough_acidity* (1.03, p = 0.087).

In SEDA, a **positive Mean drop** represents a loss of liking.

5.7.5 Global Defects Applied to Each Product

With **Consumer threshold = 20%** and **Mean-drop threshold = 1**, no row receives **Priority = Yes**.

Too_much_odor_fruit is the only globally significant defect, but it concerns only 6.38% of consumers for **284**, 4.26% for **859**, and 4.26% for **972**. It is therefore statistically associated with a liking loss across the global product space, but it is too rare within each product to exceed the practical 20% threshold.

Conversely, *Too_fluid* is very frequent — 46.81% for **284**, 70.21% for **859**, and 38.30% for **972** — but its **Global mean drop = 0.45** with p = 0.275. High prevalence and global penalty therefore do not necessarily coincide.

5.7.6 Classical product-specific penalties

The **Product-Specific Classical Penalty Analysis** provides a directly local view.

For **284**, several deviations are associated with a significant liking loss:

- *Odor_Fruit* — *Below JAR*: 46.81% of consumers, Mean drop = 1.00, 95% CI [0.04, 1.96], p = 0.042;
- *Odor_Fruit* — *Above JAR*: 6.38%, Mean drop = 2.27, 95% CI [0.47, 4.08], p = 0.016;
- *Sweetness* — *Below JAR*: 21.28%, Mean drop = 1.22, 95% CI [0.04, 2.40], p = 0.044;
- *Texture* — *Below JAR*: 57.45%, Mean drop = 1.06, 95% CI [0.10, 2.02], p = 0.032.

For *Fluidity* — *Above JAR*, 46.81% of consumers are concerned, but Mean drop is only 0.69 with p = 0.182. The deviation is therefore frequent without being clearly penalising for **284**.

For **859**, *Fluidity* — *Above JAR* gives a particularly instructive result: **70.21%** of consumers are concerned, but Mean drop is only **0.87**, p = **0.298**. Likewise, *Texture* — *Below JAR* concerns **80.85%** of consumers but has Mean drop = **0.98**, p = **0.272**. These two deviations are extremely frequent without a clearly established hedonic impact in the product-specific analysis.

The two significant effects for **859** instead concern very small groups:

- *Odor_Fruit* — *Above JAR*: 4.26%, Mean drop = 3.55, $p = 0.013$;
- *Taste_Fruit* — *Above JAR*: 6.38%, Mean drop = 2.83, $p = 0.019$.

The effects are large but rare, which limits their operational priority.

For **972**, *Fluidity* — *Above JAR* concerns 38.30% of consumers with Mean drop = 1.06 and $p = 0.080$: the signal is compatible with a liking loss but remains insufficient to conclude at the 5% level.

Two other effects are significant:

- *Color* — *Above JAR*: 6.38%, Mean drop = 2.74, $p = 0.009$;
- *Taste_Van* — *Above JAR*: 21.28%, Mean drop = 1.57, 95% CI [0.19, 2.95], $p = 0.028$.

The second is especially useful because it combines frequency above 20%, mean drop above 1, and $p < .05$. *Taste_Van* — *Above JAR* is therefore the clearest product-specific optimisation lever for **972**.

5.7.7 What the product-specific plots show

The **Product-Specific Penalty Plots** make this logic visible: the vertical line marks 20% of consumers and the horizontal line marks a mean drop of 1.

For **284**, several significant points lie in or near the upper-right region, especially lack of texture, sweetness, and fruity odor.

For **859**, *Fluidity* — *Above JAR* and *Texture* — *Below JAR* lie far to the right of the graph, meaning that they are very frequent, but remain below or around the mean-drop threshold and are not significant. The two significant effects instead have large mean drops but lie far to the left because they concern few consumers.

For **972**, *Taste_Van* — *Above JAR* combines frequency, effect size, and statistical evidence, whereas *Fluidity* — *Above JAR* is a more uncertain signal.

These plots show why frequency, effect size, and uncertainty should be considered together.

5.7.8 Multivariate synthesis of defects

The **Description of the Products According to Defects** table uses the exploratory threshold $\text{proba} = 0.5$. The displayed rows should therefore not all be described as “significant at 5%”.

Product **859** is mainly oriented toward *Texture = Not_enough_texture* (V-test = 1.92, $p = 0.055$) and *Fluidity = Too_fluid* (1.75, $p = 0.080$), while *Color = A_little_plain* (-2.05, $p = 0.040$) and *Odor_Van = Not_enough_odor_van* (-1.97, $p = 0.049$) are under-represented.

Product **972** is notably oriented toward *Sweetness = Too_much_sweetness* (V-test = 1.71, $p = 0.087$), while *Texture = Not_enough_texture* is under-represented (V-test = -1.80, $p = 0.072$).

Product **284** shows a less marked structure; its strongest positive category is *Color = A_little_plain* (V-test = 1.32, $p = 0.187$).

5.7.9 Representation of the Products and the Defects

The correspondence analysis is entirely summarised by the two available dimensions: **Dim. 1 = 74.78%** and **Dim. 2 = 25.22%**.

Dim. 1 mainly opposes **859** and **972**, while **284** occupies a more intermediate position. The map summarises the non-JAR profile structure, but product-to-defect relationships should be interpreted relative to the factor dimensions rather than from simple row-to-column proximity.

5.7.10 Synthetic interpretation

The example mainly shows that the different pieces of JAR information are not redundant.

The **global** view identifies *Too_much_odor_fruit* as the only defect clearly associated with a liking loss across the whole product space, but this defect is too rare within each product to become a priority under the 20% threshold.

The **product-specific** view instead shows that extremely frequent deviations may be weakly or non-penalising: this is the case for *Too_fluid* and *Not_enough_texture* for **859**. It also reveals local levers, including several deviations for **284** and especially *Taste_Van — Above JAR* for **972**.

What we take away from this example: useful JAR analysis is not about finding “the most penalising defect”. It distinguishes what is **frequent**, what is **globally penalising**, what is **specifically penalising for a product**, and what combines enough frequency, effect size, and statistical evidence to become an **optimisation priority**.

5.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured to use **System R** as described in Chapter 1, check **R code**. The generated script loads **SensoMineR** and **FactoMineR**, and uses the dataset currently open in jamovi as **data**.

The code successively reproduces the global multidimensional penalty with **JAR()**, the classical product-specific diagnostic, and the products $\times$ defects synthesis with **descfreq()** and **CA()**. It also carries over the thresholds and keyword patterns selected in the interface.

Graphics calls are written so that each product produces a separate output in the Rj Editor. No additional setting is required.

5.9 Practical tips

- Always inspect **JAR Coding Used for the Classical Penalty Analysis** before interpreting product-specific penalties.
- JAR variables must be defined as **Nominal** in jamovi and must contain exactly the level specified in **Coding of the JAR level**.
- An **Unclassified** category indicates a Below/Above JAR grouping problem: adapt the keyword patterns before interpreting the classical analysis.
- Do not confuse the two views: **Global Multidimensional Penalties** describes the whole product space, whereas **Product-Specific Classical Penalty Analysis** answers the question separately for each product.
- A **positive Mean drop** indicates a liking loss. Interpret its magnitude together with the percentage of consumers concerned, the confidence interval, and the p-value.
- **Consumer threshold (%)** and **Mean-drop threshold** are practical decision thresholds, not significance thresholds.
- A frequent defect is not necessarily penalising; a strongly penalising but very rare defect is not necessarily a priority.
- Use the correspondence-analysis map to understand the structure of defect profiles, not to conclude that a defect affects liking.

5.10 In summary

Use this analysis when JAR responses are available together with a liking measure and the objective is to identify deviations from the ideal level that may guide product optimisation.

SEDA first checks the **JAR coding**, then describes the **Below/JAR/Above response frequencies**, and provides two complementary views: a **global penalty analysis** across the whole product space and a **classical product-specific penalty analysis**. The multivariate

correspondence-analysis synthesis completes these diagnostics by showing how products are structured according to their non-JAR categories.

Read the coding and distributions first, then the global penalties, next the product-specific penalties, and finish with the factorial synthesis.

The main point is that an optimisation priority should not rely on a single cutoff: it should combine **defect frequency**, **mean-drop magnitude**, and **statistical uncertainty**.

6 Napping Data — Consensus Representation

6.1 Introduction

This analysis is designed for **Napping** (*projective mapping*) data, in which each subject positions a set of stimuli in a two-dimensional space according to their perceived similarities and differences. It makes it possible to construct a **consensus representation** of the stimuli from several individual maps and, if desired, to group the stimuli into **clusters**. The function relies on a **Multiple Factor Analysis (MFA)** to merge the different pairs of coordinates into a single space.

Why not use a standard alternative?

A PCA applied directly to all coordinate columns would account for relationships among the variables, but would ignore their **grouping by subject**. In Napping, the data are naturally organised as **X/Y pairs** provided by each subject. Here, each pair is treated as a **group** in the MFA, which makes it possible to construct a more suitable compromise between the different individual maps. This MFA uses `type = "c"` in FactoMineR: the coordinates are **centred but not standardised**, preserving the geometry and relative scale of each Napping sheet rather than forcing X and Y to have the same variance.

6.2 Understanding what happens under the hood

6.2.1 The data are read as pairs of coordinates

The function expects in **Active (X,Y) Coordinate Pairs** a set of numeric columns organised as consecutive pairs: X then Y for a first subject, X then Y for a second subject, and so on. The same principle applies to **Supplementary (X,Y) Coordinate Pairs**. If the number of columns is odd in either block, the analysis is rejected. The active block must contain at least **4 columns**, that is, **2 pairs**. The same coordinate cannot be selected as both active and supplementary.

6.2.2 An MFA on groups of size 2, without standardisation

The core of the analysis is an **MFA**. Each X/Y pair in **Active (X,Y) Coordinate Pairs** is treated as a group of two quantitative variables. The MFA constructs a common space that summarises the information carried by all subjects while **balancing the contribution of the groups**. With `type = "c"`, the coordinates are centred but not standardised, preserving the relative scale of the two coordinates in each individual configuration.

6.2.3 The role of supplementary coordinates

If you provide variables in **Supplementary (X,Y) Coordinate Pairs**, they are projected as **supplementary groups** in the MFA. They do not participate in the construction of the axes, but make it possible to locate other subjects in the same space.

6.2.4 Automatic description of the dimensions

Once the MFA has been computed, the function produces an **automatic description of the dimensions** filtered by **Significance threshold (%)** and **Number of dimensions**. This output helps understand which variables are most strongly linked to the first axes.

6.2.5 Saving dimensions and clustering

The **Number of dimensions to save/use for clustering** field determines how many first dimensions are saved in **Coordinates** and used for clustering. If fewer dimensions are available than requested, SEDA uses those that can actually be computed. If you activate **Representation of the clusters** or request **Cluster variable**, the function runs an **HCPC** on these retained dimensions. By default, clustering uses the **first 5 dimensions**, and **Number of clusters = -1** lets FactoMineR choose the number of clusters automatically. The percentages displayed on the HCPC map refer to this retained clustering space and may therefore differ from those on the full MFA map.

6.3 Expected data structure

The data must contain:

- one optional but strongly recommended column for **Stimuli Labels**;
- at least **two pairs** of numeric columns in **Active (X,Y) Coordinate Pairs**;
- optionally, numeric pairs in **Supplementary (X,Y) Coordinate Pairs**.

Each row corresponds to a **stimulus**. This distinguishes this analysis from other analyses where each row corresponds to an individual response. SEDA requires at least **three stimuli**. If **Stimuli Labels** is not completed, the function uses row numbers as stimulus names.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimuli Labels** is categorical if used, with no missing or empty values and one unique label per stimulus;
- all active and supplementary coordinates are numeric, complete, and finite;
- the number of columns in each coordinate block is **even**;
- no coordinate is selected as both active and supplementary;
- each X/Y pair has variability in at least one of its two coordinates.

6.4 Interface guide

The analysis appears in **SEDA > Free Perception > Analysis of Napping Data**. The interface includes three variable drop zones, an **Instructions and R Code** section, followed by the **Graphic Options**, **Numerical Indicators**, and **Clustering** menus.

6.4.1 Variables to complete

6.4.1.1 Stimuli Labels

Role: provides the names of the stimuli in the outputs. **Impact on the computation:** this variable does not modify the MFA itself, but determines the labels displayed in the graphs and the names associated with the saved coordinates. **Practical tip:** use short and unique labels; otherwise, the maps quickly become difficult to read.

6.4.1.2 Active (X,Y) Coordinate Pairs

Role: groups together the active coordinates used to construct the consensus representation. **Impact on the computation:** each X/Y pair becomes a group in the MFA. The number of columns must be even, and the function requires at least **4 columns**. **Practical tip:** strictly respect the order X, Y, X, Y... to avoid incorrectly organising the groups.

6.4.1.3 Supplementary (X,Y) Coordinate Pairs

Role: adds pairs of coordinates projected as supplementary groups. **Impact on the computation:** these coordinates do not influence the construction of the axes, but are projected into the MFA space. **Practical tip:** use this block to project a second panel without modifying the main compromise.

6.4.2 Instructions and R Code

6.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

6.4.2.2 R code

Role: displays the R script corresponding to the analysis in an **R Code** output. **Impact on the computation:** none. **Practical tip:** enable this option to reproduce the analysis in the **Rj Editor**. It is **disabled by default**.

6.4.3 Graphic Options

6.4.3.1 X-axis

Role: chooses the dimension displayed on the horizontal axis. **Impact on the computation:** it does not modify the MFA, but changes the factor plane displayed in the graphs. **Practical tip:** start with 1, the default value.

6.4.3.2 Y-axis

Role: chooses the dimension displayed on the vertical axis. **Impact on the computation:** it does not modify the MFA, but changes the factor plane displayed in the graphs. **Practical tip:** start with 2, the default value.

6.4.4 Numerical Indicators

6.4.4.1 Significance threshold (%)

Role: sets the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the automatic description will be. **Practical tip:** the default value of 5% is a good starting point.

6.4.4.2 Number of dimensions

Role: determines the number of axes described automatically. **Impact on the computation:** it controls the dimension description output and also helps determine the number of MFA dimensions computed internally. **Practical tip:** leave 2 for a first reading.

6.4.4.3 Number of dimensions to save/use for clustering

Role: defines the number of first dimensions to save and use for clustering. **Impact on the computation:** it determines the columns saved in **Coordinates** and the dimensions passed to HCPC; if fewer dimensions are available, SEDA uses the maximum that can be computed. **Practical tip:** the default value of 5 is a good starting point.

6.4.4.4 Coordinates

Role: saves the factorial coordinates of the stimuli in the jamovi data sheet. **Impact on the computation:** it does not add a new calculation, but saves the results of the MFA. **Practical tip:** activate this output if you want to reuse the dimensions in other jamovi analyses.

6.4.5 Clustering

6.4.5.1 Representation of the clusters

Role: activates the display of the clustering graph. **Impact on the computation:** when this option is checked, the function computes the HCPC classification if necessary and displays the map of stimuli according to clusters. **Practical tip:** leave it unchecked for a first reading, then activate it to complement the consensus map with a group-based reading.

6.4.5.2 Number of clusters

Role: sets the number of clusters to extract. **Impact on the computation:** with -1, the number of clusters is determined automatically; otherwise, you can impose an integer between 2 and the number of stimuli minus 1. **Practical tip:** start with -1, then set a number only if you have a clear analytical justification.

6.4.5.3 Cluster variable

Role: saves the cluster membership of each stimulus as a new variable in jamovi. **Impact on the computation:** it does not add a new calculation, but saves the results of the classification. **Practical tip:** useful if you then want to cross the clusters with other variables in jamovi.

6.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the coordinate columns are numeric and organised in X/Y pairs.
3. If you have a stimulus identifier column, check that it is categorical.
4. Open **SEDA > Free Perception > Analysis of Napping Data**.
5. Place the stimulus name column in **Stimuli Labels**.
6. Place all active pairs in **Active (X,Y) Coordinate Pairs**.
7. If needed, place the supplementary pairs in **Supplementary (X,Y) Coordinate Pairs**.
8. Check that the coordinates are complete and that each block contains an even number of columns.
9. Leave **X-axis = 1** and **Y-axis = 2**.
10. Leave **Significance threshold (%) = 5**, **Number of dimensions = 2**, and **Number of dimensions to save/use for clustering = 5**.
11. Start with **Eigenvalue Decomposition**, then read **Representation of the Stimuli** and **Representation of the Subjects**.
12. Then consult **Automatic Description of the Dimensions**.
13. Activate **Representation of the clusters** if you want to display **Representation of the Stimuli According to Clusters**.
14. Check **R code** to obtain the reproducible script, and optionally save **Coordinates** and **Cluster variable**.

6.6 Interpretation of results

6.6.1 Representation of the Stimuli

This is the important result of the function. It represents the stimuli in the consensus factor space constructed by the MFA. Two close stimuli were positioned similarly by the set of subjects; two distant stimuli are perceived as more different. The percentage displayed on each axis indicates the share of variance explained by the corresponding dimension.

Element	Meaning
Two close stimuli	Similar global perceptions
Two distant stimuli	Different global perceptions
Compact group of stimuli	Family of stimuli perceived as close

Point of vigilance: the stimulus map shows a perceptual structure, but it is not always sufficient to name the axes. Adding supplementary information, such as verbalisation, can greatly help interpretation.

6.6.2 Representation of the Subjects

This map represents the X/Y coordinate groups — that is, the subjects' individual configurations — in the factor space. It shows how each Napping sheet relates to the dimensions of the compromise.

Element	Meaning
Groups occupying similar positions	Configurations related similarly to the common dimensions
Group far from the origin on one axis	Configuration strongly related to that dimension
Group close to the origin	Configuration weakly related to the displayed dimensions

6.6.3 Eigenvalue Decomposition — eigenvalue table

Element	Meaning
Dim.	Dimension number
Eigenvalue	Importance of the dimension
% of variance	Share of variability explained

Element	Meaning
Cumulative %	Cumulative share explained

If the first two dimensions explain a large share of the variance, the 1/2 plane provides a good summary; otherwise, it may be useful to explore other factor planes.

6.6.4 Automatic Description of the Dimensions

This output automatically describes the retained dimensions according to **Significance threshold (%)** and **Number of dimensions**. It helps identify which variables are most strongly linked to the axes and becomes particularly useful when **Supplementary (X,Y) Coordinate Pairs** have been added.

6.6.5 Representation of the Stimuli According to Clusters (*available if “Representation of the clusters” is activated*)

This graph shows the stimuli in the factor space, grouped according to the clusters derived from the HCPC classification. Classification uses the first dimensions defined by **Number of dimensions to save/use for clustering**, not only the two displayed axes. The percentages on the HCPC map are therefore computed relative to this retained space.

Element	Meaning
Stimuli in the same cluster	Similar positioning profiles
Clearly separated clusters	Well-differentiated families of stimuli

6.6.6 Coordinates (*data output*)

This output saves the factorial coordinates of the stimuli on the first dimensions retained by **Number of dimensions to save/use for clustering**. Each column corresponds to one dimension (Dim. 1, Dim. 2, etc.).

6.6.7 Cluster variable (*data output*)

This output saves the cluster membership of each stimulus as a nominal variable. It is useful for subsequent analyses in jamovi.

6.7 Example: the perfumes_napping dataset

The **perfumes_napping** dataset describes 12 perfumes positioned on a sheet (with dimensions 60 by 40) by 60 subjects, that is, 120 coordinate columns (one X/Y pair per subject). It is used to illustrate the function outputs. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

6.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimuli Labels	variable Ident
Active (X,Y) Coordinate Pairs	the first 60 coordinate columns (30 pairs)
Supplementary (X,Y) Coordinate Pairs	the following 60 coordinate columns (30 pairs)
X-axis / Y-axis	1 / 2
Significance threshold (%)	5
Number of dimensions	2
Number of dimensions to save/use for clustering	5
Representation of the clusters	checked
Number of clusters	-1
R code	checked

6.7.2 Analysis of the *Representation of the Stimuli* graph

The map reveals a fairly clear perceptual structure on the Dim. 1 (25.07%) / Dim. 2 (18.03%) plane. On the horizontal axis, **Aromatics Elixir** and **Shalimar** are clearly located on the right (positive coordinates on Dim. 1), whereas **J'adore EP**, **J'adore ET**, **Coco Mademoiselle**, **Cinéma**, **Pleasures**, and **Pure Poison** are located on the left. **Chanel N5** differs from this right-hand group by a low position on the second dimension (strongly negative value on Dim. 2), which slightly isolates it from the Shalimar / Aromatics Elixir pair. On the vertical axis, **Angel**, **Lolita Lempicka**, and **L'Instant** are clearly located above the centre, forming a coherent perceptual group at the top of the plane.

6.7.3 Analysis of the *Representation of the Subjects* graph

The subject map represents the 60 coordinate groups, S1 to S60. Groups far from the origin on Dim. 1, such as **S1** or **S20**, are strongly related to the first common dimension. Groups such as **S4** or **S30**, located higher on Dim. 2, are more strongly related to the second. The many groups close to the origin are less strongly associated with the two displayed dimensions. This map therefore describes the relationship between individual configurations and the compromise; it is not, by itself, a numerical index of consensus.

6.7.4 Analysis of the *Eigenvalue Decomposition* table

The first two dimensions explain 43.09% of the total variance (Dim. 1: 25.07%, Dim. 2: 18.03%). The 1/2 plane therefore provides a useful first summary, but more than half of the variability lies beyond this plane. The following dimensions remain non-negligible (Dim. 3: 12.31%, Dim. 4: 9.58%, Dim. 5: 7.96%), which justifies examining other factor planes when needed.

6.7.5 Analysis of the *Automatic Description of the Dimensions* table

The **Automatic Description of the Dimensions** output identifies the X/Y coordinates most correlated with each axis. For **Dim. 1**, the strongest positive correlations are **X20** ($r = 0.93$), **X1** ($r = 0.92$), and **Y21** ($r = 0.90$). At the opposite end, **Y48** ($r = -0.91$) and **X42** ($r = -0.85$) have the strongest negative correlations. These results identify which directions of the individual configurations are most aligned with, or opposed to, the first compromise dimension. They do not by themselves provide a sensory meaning for the axis.

6.7.6 Analysis of the *Representation of the Stimuli According to Clusters* graph

With **Number of dimensions to save/use for clustering = 5**, the retained MFA dimensions account for 72.95% of the total inertia. The HCPC map therefore expresses Dim. 1 and Dim. 2 relative to this five-dimensional space: approximately **34.37%** and **24.72%**, rather than 25.07% and 18.03% on the full MFA map.

With **Number of clusters = -1**, the automatic classification retains **three clusters**:

- **Cluster 1** (black): **Coco Mademoiselle**, **J'adore EP**, **J'adore ET**, **Cinéma**, **Pleasures**, and **Pure Poison**;
- **Cluster 2** (red): **Angel**, **Lolita Lempicka**, and **L'instant**;
- **Cluster 3** (green): **Shalimar**, **Aromatics Elixir**, and **Chanel N5**.

On the 1/2 plane, the first group lies mainly on the left, the second at the top, and the third on the right and lower part of the map. However, the classification is calculated from the **first five retained dimensions**, not only from this plane. Without external sensory information or verbalisation, qualities such as “floral”, “light”, or “intense” should not be assigned automatically to these clusters.

6.7.7 Synthetic interpretation

This example clearly shows the value of the function. It does not only provide an average map of the perfumes: it constructs a **consensus space**, shows how the individual configurations relate to its dimensions, and identifies which **groups of perfumes** emerge from the classification. The three clusters are clear in this example, but their sensory interpretation would require additional information.

What we take away from this example: Napping analysis transforms a set of individual maps into a consensus representation, then makes it possible to examine the relationships between individual configurations and common dimensions as well as the groups of stimuli.

6.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured with **System R** as explained in Chapter 1, check **R code** to display the script corresponding to the analysis. The dataset currently open in jamovi is available as **data**.

For Napping, the script loads **FactoMineR**, constructs one X/Y pair per subject, and runs **MFA()** with **type = "c"**. Supplementary pairs are specified through **num.group.sup**. The script then reproduces the eigenvalues, the stimulus and subject maps with **plot.MFA()**, and the automatic dimension description with **dimdesc()**.

If clustering is requested, the code retains exactly the first dimensions defined by **Number of dimensions to save/use for clustering**, runs **HCPC()** on that retained space, and displays the map with **plot.HCPC()**. In this example, the generated code uses 30 active subjects, 30 supplementary subjects, and 5 dimensions for clustering.

6.9 Practical tips

- Always check that the columns in **Active (X,Y) Coordinate Pairs** are complete and supplied in X/Y pairs.
- Use **Stimuli Labels** almost systematically and check that the labels are unique.

- Start with **X-axis = 1** and **Y-axis = 2**; if the first two dimensions summarise little variance, explore other factor planes.
- Keep **Number of dimensions to save/use for clustering = 5** for a first trial, especially if you are considering clustering.
- Leave **Number of clusters = -1** at first to let the function choose automatically.
- Read **Representation of the Subjects** as the relationship between individual configurations and the common dimensions, not as a direct consensus index.
- Do not assign sensory meaning to axes or clusters without verbalisation or external information that can characterise them.
- Activate **Coordinates** and **Cluster variable** if you want to reuse the results in other jamovi analyses.

6.10 In summary

This analysis constructs a **consensus map of the stimuli** from Napping data by combining the individual maps through a **centred but non-standardised MFA** that balances the contribution of the groups. It represents the structure of the stimuli and the relationship between individual configurations and the common dimensions, and can be complemented by an **HCPC clustering** of the stimuli on the retained dimensions. Its main value is to transform a set of individual maps into a common representation of perceptual proximities, without requiring a predefined list of attributes.

7 Sorting Data — Analysis of Categorisation

7.1 Introduction

This analysis is designed for **free sorting** (*sorting task*) data, in which each subject groups stimuli into categories according to their own perception. It makes it possible to obtain a **multivariate representation of the stimuli** from the categorisations provided by the subjects, and, if needed, an **automatic classification** of the stimuli into groups. When the groups have been verbalised, the function also provides a **textual description of the stimuli** based on the words used to name these groups.

Why not use a standard alternative?

A classical approach consists of transforming the sorting data into a similarity matrix between stimuli, then applying a projection method such as MDS. This strategy is useful, but it removes part of the individual information carried by each subject. Here, the function directly analyses the columns in **Active Subjects** as qualitative variables and applies a **Multiple Correspondence Analysis (MCA)**, which preserves more of the structure of individual responses and makes it possible to represent stimuli and categories simultaneously in the same space.

7.2 Understanding what happens under the hood

7.2.1 The stimuli are the individuals in the analysis

The function reads the columns placed in **Active Subjects** as categorical variables. Each row corresponds to a **stimulus**, and each column indicates in which group a subject placed this stimulus. If **Stimuli Labels** is completed, this variable is used to name the stimuli in the outputs; otherwise, the function uses row numbers.

7.2.2 An MCA on the categorisations provided by the subjects

The core of the function is an **MCA** applied to the columns in **Active Subjects**. In this framework, two stimuli are close when their sorting profiles are similar across the active subjects. The columns placed in **Supplementary Subjects**, if any, are projected as supplementary

qualitative variables: they appear in the factor space, but do not participate in the construction of the axes.

7.2.3 The role of Ventilation level (%)

The **Ventilation level (%)** parameter is passed to FactoMineR's MCA. Categories whose frequency falls below the threshold are ventilated before the MCA: the affected stimuli are randomly reassigned to other observed categories of the same variable. This limits the influence of extremely rare categories but can introduce slight variation between runs. The default value is 5%; a value of 0% disables ventilation.

7.2.4 Textual description of the stimuli

The function constructs a stimuli $\times$ words correspondence table from the group labels, then applies a frequency description procedure. This part is used to identify, for each stimulus, the most characteristic words among the labels used by the subjects. It becomes particularly useful when the groups have been named with interpretable sensory terms.

7.2.5 Automatic description of the dimensions and clustering

Once the MCA has been computed, the function produces an **automatic description of the dimensions** according to **Significance threshold (%)** and **Number of dimensions**. If you activate **Representation of the clusters**, the function applies **HCPC** to the stimulus coordinates on the retained components. By default, clustering is based on the **first 5 components**, and **Number of clusters = -1** lets HCPC choose the number of groups automatically. The percentages displayed on the HCPC map refer to this retained space and may therefore differ from those on the full MCA map.

7.3 Expected data structure

The data must contain:

- one optional column for **Stimuli Labels**;
- at least **two** categorical columns in **Active Subjects**;
- optionally, categorical columns in **Supplementary Subjects**.

Each row corresponds to a **stimulus**. This point is important: unlike in other analyses, here a row does not correspond to an individual response, but to a stimulus described by the categorisations provided by several subjects.

Before running the analysis in jamovi, check in the **Data** tab that:

- **Stimuli Labels** is categorical if you use it;
- all columns in **Active Subjects** are categorical (factors);
- all columns in **Supplementary Subjects** are categorical;
- you have placed at least **two** variables in **Active Subjects**.

7.4 Interface guide

The analysis appears in **SEDA > Free Perception > Analysis of Sorting Data**. The interface includes three variable drop zones, the **Instructions and R Code** section, and then the **Random Assignment for Rare Categories**, **Graphic Options**, **Numerical Indicators**, and **Clustering** sections.

7.4.1 Variables to complete

7.4.1.1 Stimuli Labels

Role: provides the names of the stimuli in the outputs. **Impact on the computation:** it does not modify the MCA, but determines the labels displayed in the graphs and the names associated with the saved coordinates. **Practical tip:** use short and unique labels; otherwise, the maps become difficult to read.

7.4.1.2 Active Subjects

Role: groups together the sorting variables that construct the factor space, one per subject. **Impact on the computation:** these are the active variables of the MCA. Clustering then uses the stimulus coordinates produced by this MCA. At least **two** variables are required. **Practical tip:** check that each column is categorical in jamovi.

7.4.1.3 Supplementary Subjects

Role: groups together variables projected as supplementary qualitative variables. **Impact on the computation:** they do not influence the construction of the axes, but their position in the space can help interpretation. **Practical tip:** useful for projecting a second group of subjects without modifying the main result.

7.4.2 Instructions and R Code

7.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** leave it checked during the first uses. It is **enabled by default**.

7.4.2.2 R code

Role: displays the R script corresponding to the current analysis. **Impact on the computation:** none. **Practical tip:** check this option if you want to reproduce the analysis in the **Rj Editor**; it is disabled by default.

7.4.3 Random Assignment for Rare Categories: Ventilation level (%)

Role: sets the threshold below which a category is ventilated before the MCA. **Impact on the computation:** stimuli belonging to a rare category are randomly reassigned to other observed categories of the same variable. The default value is 5%. **Practical tip:** leave 5% for a first trial; use 0% if you want to keep rare categories without ventilation.

7.4.4 Graphic Options (*collapsed by default*)

7.4.4.1 X-axis and Y-axis

Role: choose the dimensions displayed on the horizontal and vertical axes. **Impact on the computation:** they do not modify the MCA, but change the displayed factor plane. **Practical tip:** start with **X-axis = 1** and **Y-axis = 2**, the default values.

7.4.4.2 Active categories

Role: controls the display of active categories on the category map. **Impact on the computation:** no effect on the MCA; it only affects visibility in the graph. **Practical tip:** leave it checked at first. Uncheck it if the map is too crowded.

7.4.4.3 Selection of categories

Role: filters the displayed categories according to a quality-of-representation criterion. **Impact on the computation:** no effect on the MCA; it only affects display. **Practical tip:** the default value **cos2 10** is a good compromise for lightening the map without losing the best-represented categories.

7.4.4.4 Supplementary categories

Role: controls the display of categories derived from **Supplementary Subjects**. **Impact on the computation:** no effect on the MCA. **Practical tip:** leave it checked if you use supplementary variables and want to compare them visually with the active categories.

7.4.5 Numerical Indicators (*collapsed by default*)

7.4.5.1 Significance threshold (%)

Role: sets the significance threshold for the automatic description of the dimensions. **Impact on the computation:** the stricter the threshold, the more selective the output. **Practical tip:** the default value of 5% is well suited to a first reading.

7.4.5.2 Number of dimensions

Role: determines the number of axes described automatically. **Impact on the computation:** it controls the dimension description output. **Practical tip:** leave 2 for a first trial.

7.4.5.3 Description of the Stimuli from Sorting Labels — Significance threshold (%)

Role: sets the threshold used for the textual description of the stimuli. **Impact on the computation:** a stricter threshold produces a shorter description; a broader threshold retains more words. **Practical tip:** leave 5% at first. If the textual description is too sparse, test a slightly broader threshold.

7.4.5.4 Long-format textual data

Role: displays the intermediate textual dataset in long format. **Impact on the computation:** it does not add a new calculation; it makes it possible to check how the labels were reorganised before the textual analysis. **Practical tip:** activate this option if you want to inspect the transformation of the verbal data.

7.4.5.5 Stimulus Table Options — Coordinates, Contributions, Cosine

Role: activate the display of the numerical tables of coordinates, contributions, and cosines for the stimuli. **Impact on the computation:** no effect on the MCA; they add numerical result tables. **Practical tip:** disabled by default. Activate them if you want precise values for complementary analyses.

7.4.5.6 Subject Table Options — Coordinates, Contributions, Cosine

Role: same principle for the MCA subjects. **Impact on the computation:** no effect on the MCA. **Practical tip:** useful if you want to examine their positions and quality of representation precisely.

7.4.5.7 Number of components (*Save*)

Role: defines the number of components saved in **Coordinates** and used for classification. **Impact on the computation:** HCPC works on the stimulus coordinates in this retained space. **Practical tip:** the default value of 5 is a useful starting point.

7.4.5.8 Coordinates (*Save*)

Role: saves the factorial coordinates of the stimuli in the jamovi data sheet. **Impact on the computation:** no effect on the MCA. **Practical tip:** activate it if you want to reuse the components in other jamovi analyses.

7.4.6 Clustering (*collapsed by default*)

7.4.6.1 Representation of the clusters

Role: activates HCPC classification and displays the corresponding graph. **Impact on the computation:** HCPC uses the first components defined by **Number of components**. It is unchecked by default. **Practical tip:** leave it unchecked for a first reading, then activate it to complement the factor map with a group-based reading.

7.4.6.2 Number of clusters

Role: sets the number of clusters to extract. **Impact on the computation:** with -1, the number is chosen automatically; any positive value imposes that number. **Practical tip:** start with -1, then set a number only if you have a clear analytical reason.

7.4.6.3 Cluster variable (Save)

Role: saves the cluster membership of each stimulus in the jamovi data sheet. **Impact on the computation:** it does not add a new calculation, but saves the classification results. **Practical tip:** useful if you then want to cross the clusters with other variables in jamovi.

7.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that all columns in **Active Subjects** are categorical and that you have at least two of them.
3. Check that **Stimuli Labels** is categorical if you use it.
4. Open **SEDA > Free Perception > Analysis of Sorting Data**.
5. Place the stimulus name variable in **Stimuli Labels**.
6. Place the sorting variables that construct the space in **Active Subjects**.
7. If needed, place other sorting variables in **Supplementary Subjects**.
8. Leave **Ventilation level (%) = 5**, **X-axis = 1**, **Y-axis = 2**, both **Significance threshold (%) = 5**, **Number of dimensions = 2**, and **Number of components = 5** for a first trial.
9. If subjects verbalised their groups, start with **Words Associated with the Stimuli**.
10. Then read **Eigenvalue Decomposition**, **Representation of the Stimuli**, **Representation of the Subjects**, and **Representation of the Sorting Categories**.
11. Consult **Automatic Description of the Dimensions** to identify the subjects and categories associated with the axes.
12. Activate **Representation of the clusters** if you want an HCPC classification.

7.6 Interpretation of results

7.6.1 Description of the Stimuli from Sorting Labels — Words Associated with the Stimuli

This table describes each stimulus based on the words used in the group labels. It is truly informative only if the groups were named with interpretable words.

Element	Meaning
<i>First column</i>	Stimulus concerned
Word	Characteristic word
Internal %	Percentage of the word for this stimulus
Global %	Global percentage of the word

Element	Meaning
Internal frequency	Frequency of the word for this stimulus
Global frequency	Global frequency of the word
p	P-value of the test
V-test	Standardised intensity of the association

- A word with an **Internal %** clearly higher than **Global %** characterises this stimulus more strongly.
- A low **p-value** indicates a clearer association.
- A positive **V-test** indicates over-representation of the word for this stimulus; a negative **V-test** indicates under-representation.

7.6.2 Representation of the Stimuli — stimulus map

This is the central result of the function. It represents the stimuli in the MCA factor space.

Element	Meaning
Two close stimuli	Their sorting profiles are similar across the active subjects
Two distant stimuli	Their sorting profiles are more different
Compact group of stimuli	Family of stimuli with similar sorting profiles

The percentage displayed on each axis indicates the share of MCA inertia explained by that dimension. Without verbalisations or external information, the axes describe **sorting contrasts** but do not automatically have a sensory meaning.

7.6.3 Representation of the Sorting Categories — category map

This map represents the categories specific to each subject. Their direction from the origin helps characterise the sides of the axes; the direct distance between a category and a stimulus should not be interpreted as an ordinary measure of similarity. The **Active categories**, **Supplementary categories**, and **Selection of categories** options control what is displayed.

Point of vigilance: with many subjects, this map can become very crowded. The default value **cos2 10** helps keep only a more readable subset of categories.

7.6.4 Representation of the Subjects — subject map

This map represents the subjects as the qualitative variables of the MCA. It shows which subjects are associated with the common dimensions. The **Subjects Associated with the Dimensions** table quantifies this association through R^2 and p-values; the map should not be read as a direct index of consensus between subjects.

7.6.5 Eigenvalue Decomposition — eigenvalue table

Element	Meaning
Dim.	Dimension number
Eigenvalue	Eigenvalue
% of variance	Share explained by the dimension
Cumulative %	Cumulative share explained

If the first two dimensions explain a large share of the variance, the 1/2 plane summarises the structure of the sorting data well. Otherwise, explore other factor planes.

7.6.6 Automatic Description of the Dimensions

This output contains the **Subjects Associated with the Dimensions** and **Sorting Categories Associated with the Dimensions** tables. It identifies the subjects and categories associated with each dimension according to **Significance threshold (%)** and **Number of dimensions**. It therefore describes sorting contrasts; a sensory meaning can be assigned to an axis only when group labels or other external information support that interpretation.

7.6.7 Representation of the Stimuli According to Clusters (*available if “Representation of the clusters” is activated*)

This graph projects the groups obtained by **HCPC**. Classification uses the first components defined by **Number of components**, not only the two displayed axes. The percentages shown on the HCPC map are therefore computed relative to this retained space and may be higher than those on the full MCA map.

Element	Meaning
Stimuli in the same cluster	Similar sorting profiles in the retained space
Clearly separated clusters	More differentiated families of stimuli

7.6.8 Numerical tables (*optional*)

The coordinate, contribution, and cosine tables for **stimuli** and **subjects** appear only if you check the corresponding options in **Numerical Indicators**. They provide precise values to complement the visual interpretation.

7.7 Example: the perfumes_sorting dataset

The **perfumes_sorting** dataset concerns 12 perfumes grouped by **58 subjects** during a free sorting task. It is used to illustrate the function outputs. Some groups were also verbalised, which allows the function to produce both a factorial representation of the perfumes and a textual description of the profiles. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

7.7.1 Example settings

To reproduce this example, use the following values.

Option	Value
Stimuli Labels	Product
Active Subjects	S1 to S50
Supplementary Subjects	S51 to S58
R code	checked
Ventilation level (%)	5
X-axis / Y-axis	1 / 2
Significance threshold (%) (<i>dimension description</i>)	5
Number of dimensions	2
Significance threshold (%) (<i>sorting-label description</i>)	5
Number of components	5
Representation of the clusters	checked to visualise the classification
Number of clusters	-1

Reproducibility note: with **Ventilation level (%) = 5**, rare categories may be ventilated by random reassignment before the MCA. Small numerical differences can therefore occur between runs. Use **Ventilation level (%) = 0** only if exact run-to-run reproducibility of the factor solution is required, bearing in mind that this changes the analysis.

7.7.2 Analysis of the *Description of the Stimuli* table

The table shows that several perfumes have very marked lexical profiles. A few particularly clear readings, with the actual values from the analysis:

Aromatics Elixir is strongly associated with *strong* (test value 4.08, $p < .001$; 14.16% of internal occurrences versus 4.65% overall), *ammonia* (test value 2.26), and *spicy* (2.18). Conversely, it is clearly under-represented by *fruit* (test value -2.63), *flower* (-2.40), and *sweet* (-2.10): it is the most intense and least floral or sweet perfume in the panel.

Chanel N5 stands out with *soap* (test value 3.72, $p < .001$; 7.69% versus 1.61% overall), *deodorant* (2.06), and *unpleasant* (1.99), while being under-represented by *fruit* (-2.47) and *flower* (-2.21).

Lolita Lempicka is characterised by *sweet* (test value 3.94, $p < .001$; 14.55% versus 4.98% overall) and *vanilla* (2.09), but is under-represented by *flower* (-2.34).

Pleasures is the perfume most associated with *flower* (test value 3.46, $p < .001$; 15.00% versus 5.70% overall) and *fresh* (2.05), and clearly opposed to *strong* (-2.46).

Shalimar is characterised by *old* (test value 2.67, $p = 0.008$) and *sun* (2.40), and under-represented by *flower* (-2.42).

Angel is associated with *vanilla* (2.23), *heady* (1.99), and *too* (1.99), but under-represented by *fresh* (-2.03) and *flower* (-2.19).

7.7.3 Analysis of the *Representation of the Stimuli* graph

The map reveals a readable perceptual structure on the Dim. 1 (15.53%) / Dim. 2 (14.14%) plane. On the right side of the plane, *Shalimar*, *Aromatics Elixir*, and *Chanel N5* form a group — confirming their strong, spicy, and low-floral profile. On the lower left, *J'adore ET*, *J'adore EP*, and *Pleasures* group together around a floral and fresh profile. At the top, *Angel* and *Lolita Lempicka* occupy a clearly distinct position on the second dimension — a vanilla and sweet profile. *Cinéma*, *L'instant*, and *Coco Mademoiselle* form an intermediate central-left area, with profiles that are more difficult to name without the textual description.

7.7.4 Analysis of the *Eigenvalue Decomposition* table

The first two dimensions explain 29.67% of the total variance (15.53% + 14.14%). This is clearly less concentrated than in a classical PCA: in free sorting, subjects have very different ways of structuring the space, which spreads the information across many dimensions. The 1/2 plane is a useful but partial summary — as confirmed by the following dimensions (Dim. 3: 11.03%, Dim. 4: 9.97%, etc.).

7.7.5 Analysis of the *Automatic Description of the Dimensions* table

On **Dim. 1**, the categories most strongly associated with the positive side (right, strong perfumes) are *solvent* (S52, estimate 1.31), *unpleasant* (S46, 1.27), *spicy* (S40, 1.24), *very;strong;perfume;woman;old* (S15, 1.32), *soap;ancient* (S48, 1.26), and *acid* (S54, 1.22). The negative side is associated with *flower;fresh* (S43, -0.60), *fresh;light* (S23, -0.81), and *flower* (S47, -0.68). Dim. 1 therefore clearly opposes **strong, spicy, acidic, or unpleasant** perfumes to **fresh and floral** perfumes.

On **Dim. 2**, the positive side is dominated by *vanilla;spicy;hot* (S16, 1.51), *woman;mature* (S7, 1.55), *sweet;too;strong* (S11, 1.43), *teenage;likeable* (S51, 1.54), and *vanilla;sweet* (S34, 1.05). The negative side includes *soap* (S12, -1.03) and *fresh* (S10, -0.50). Dim. 2 therefore opposes **vanilla, sweet, enveloping** perfumes to **fresher or soapy** profiles.

7.7.6 Analysis of the *Representation of the Stimuli According to Clusters* graph

With **Number of components = 5**, HCPC works on the first five MCA dimensions, which together account for **59.59%** of the inertia. The classification map therefore displays **26.06%** on Dim. 1 and **23.73%** on Dim. 2 instead of **15.53%** and **14.14%** on the MCA map: the percentages are renormalized within the five-dimensional space actually used for clustering.

With **Number of clusters = -1**, the automatic classification retains **six clusters**. The labels in parentheses below are supported by the textual and dimension descriptions, not by clustering alone:

- a cluster grouping **J'adore ET, J'adore EP, and Pleasures** (floral, fresh);
- a cluster grouping **Coco Mademoiselle, L'Instant, and Cinéma** (intermediate profiles);
- a cluster containing only **Pure Poison** (singular profile);
- a cluster grouping **Angel and Lolita Lempicka** (vanilla, sweet);
- a cluster containing only **Chanel N5** (soapy, unpleasant);
- a cluster grouping **Shalimar and Aromatics Elixir** (strong, spicy, old).

7.7.7 Synthetic interpretation

This example clearly shows the value of the function. It does not only provide a proximity map between perfumes: it also shows **which words** subjects associate with each perfume, **which oppositions structure the space**, and **which groups of perfumes** emerge from the classification. Cross-reading the description table, the factor map, and the classification makes it possible to name the families more precisely than with the map alone.

What we take away from this example: free sorting analysis makes it possible to link the factorial structure of the categorisations, their verbal interpretation, and the formation of coherent families of stimuli.

7.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured with **System R** as described in Chapter 1, check **R code** to display the script corresponding to the analysis. The dataset open in jamovi is available as **data**.

For free sorting, the script loads **FactoMineR**, builds the active and supplementary subject table, and runs **MCA()** with supplementary subjects in **quali.sup** and **Ventilation level (%)** in **level.ventil**. When verbal sorting labels are available, it also builds the textual data, uses **textual()** followed by **descfreq()**, and reproduces the eigenvalues, the maps with **plot.MCA()**, and the automatic description with **dimdesc()**.

If clustering is requested, the script extracts exactly the first coordinates defined by **Number of components**, passes them to **HCPC()**, and displays the map with **plot.HCPC()**. The percentages on this map therefore describe the retained clustering space; in the example, five dimensions account for 59.59% of the inertia, leading to 26.06% and 23.73% on the HCPC map.

7.9 Practical tips

- Always check that all columns in **Active Subjects** are categorical before running the analysis.
- Use **Stimuli Labels** almost systematically to obtain readable maps.
- Start with **Ventilation level (%) = 5**.
- If the category map is too crowded, keep **Selection of categories = cos2 10** and uncheck **Active categories** if necessary.
- Leave **Number of clusters = -1** for automatic determination at first.
- Keep **Number of components = 5** for a first clustering.
- If the first two dimensions explain little variance, explore other factor planes.
- Activate **Coordinates** and **Cluster variable** if you want to reuse the results in other jamovi analyses.

7.10 In summary

This analysis transforms free sorting data into a reading of the stimuli that is simultaneously **factorial**, **textual**, and **typological**. It combines an **MCA** on the categorisations provided

by the subjects, a **description of the stimuli based on the labels**, and, if desired, an **automatic classification** of the stimuli. Its main value is to preserve subjects' individual information while constructing a consensus and directly interpretable view of the perceptual space of the stimuli.

8 Hedonic Data — Preference Mapping

8.1 Introduction

This analysis makes it possible to construct a **preference map** from two types of information: an already computed **two-dimensional representation space**, and several **Liking Variables** corresponding to liking scores. It is used to identify which areas of this space are associated with stronger preference, and to represent **the proportion of consumers for whom each region of the plane is predicted to be favourable under the selected threshold**. In SEDA, this principle can be used both for **external preference mapping** (sensory space) and for **internal preference mapping** (hedonic space), depending on the origin of the two axes used. The analysis can also **classify consumers** from their liking profiles and produce a preference map for each segment.

Why not use a standard alternative?

A mean liking score by product makes it possible to compare products with one another, but it does not make it possible to relate preferences to an already constructed representation space. Here, the function fits a preference model **for each liking variable**, then combines these models into a **surface** indicating the **% of consumers** associated with each area of the plane. This provides a spatial reading of preferences, which is more informative than a simple table of means.

8.2 Understanding what happens under the hood

8.2.1 The function does not construct the representation space itself

The function does not calculate the axes of the plane: it expects you to provide one variable for **X-axis** and one variable for **Y-axis**. The integrated help text explicitly specifies the recommended workflow: for an external map, these axes may come from a PCA on sensory attributes; for an internal map, they may come from a PCA on hedonic scores. In both cases, the two components must first have been **saved** in the data, then selected in this module.

How to retrieve dimensions from the MEDA module

1. Open the **MEDA** module (menu **MEDA > PCA**).

2. Place your variables in the appropriate fields: for an external map, use your sensory attributes; for an internal map, use your hedonic variables.
3. Expand the **Numerical Indicators** section, then the **Save** subsection.
4. In the **Number of components** field, enter **2** (or more if you want to explore other planes).
5. Activate the **Coordinates** output: jamovi then automatically creates two new columns in your data sheet, named by default **Dim. 1** and **Dim. 2**, containing the coordinates of each stimulus on the first two components.
6. Then return to **SEDA > Hedonic Data > Preference Mapping** and place **Dim. 1** in **X-axis** and **Dim. 2** in **Y-axis**.

If your dataset contains the sensory attributes, the hedonic scores, and the stimulus identifiers in the same file — as is the case for the `senso_hedo_cocktail` dataset — this workflow can be performed entirely in jamovi without any intermediate export or import.

8.2.2 A model is fitted for each liking variable

The function first gathers the two coordinates of the plane and all columns placed in **Liking Variables**. Each liking column represents one consumer. A separate preference model is fitted for each consumer. The model depends on **Regression model**:

- **Quadratic** (default): linear effects, quadratic terms, and interaction between the two axes. At least **7 stimuli** are required for a non-saturated fit; in practice, more products are preferable.
- **Vector**: linear effects only. At least **4 stimuli** are required.
- **Circular**: linear effects with a common radial quadratic curvature. At least **5 stimuli** are required.
- **Elliptical**: linear effects and separate quadratic terms on each axis, without interaction. At least **6 stimuli** are required.

A simple interpretation is that **Vector** describes a preference direction; **Quadratic** allows the most flexible surface; **Circular** and **Elliptical** describe local optima under different shape constraints. If the geometry of the two coordinates cannot support the selected model, SEDA asks for a simpler model or a richer product space.

8.2.3 Construction of the preference surface

For each consumer, SEDA fits the selected model to the observed products and predicts liking over a regular grid covering the **X-axis** / **Y-axis** plane. Each consumer's predicted surface is then standardised to mean 0 and standard deviation 1 over the grid.

At each grid location, SEDA counts how many consumers have a standardised predicted value above **Preference threshold (SD)**. This count is converted to a percentage and forms the **% of consumers** surface.

8.2.4 Preference threshold (SD)

The threshold is expressed in standard deviations of each consumer's own predicted surface. With the default value **0**, a location is considered favourable for a consumer when predicted liking is above that consumer's average predicted level over the grid. A positive value makes the criterion more selective; a negative value broadens the favourable region. This is a **preference criterion**, not a statistical significance threshold.

8.2.5 What the final map really represents

The final map does not directly display the observed product-by-product scores. It shows a **modelled surface** constructed from the individual consumer models. The regions indicate the proportion of consumers for whom each location in the plane exceeds the selected preference threshold. The **% of consumers** legend should therefore be read as the percentage of consumers for whom a location is predicted to be favourable, not as a mean liking score.

8.2.6 Consumer classification

If **Classification of consumers** is enabled, classification is computed separately from the preference map. SEDA centres each consumer's liking profile across stimuli, computes distances between consumers, and applies Ward hierarchical clustering (**ward.D2**). Centring removes each consumer's average rating level, so the groups mainly reflect **different patterns of preference across products**, rather than a simple distinction between consumers who rate high or low overall.

With **Number of clusters = -1**, SEDA automatically selects a cut from the largest upper-level jump in the hierarchy. A fixed number of groups can also be imposed. Cluster-specific preference maps are then recomputed using the consumers in each group while keeping exactly the same product space, regression model, and preference threshold as the overall map.

8.3 Expected data structure

The data must contain:

- one optional column for **Stimuli Labels**;
- one numeric column for **X-axis**;

- one numeric column for **Y-axis**;
- at least **two** numeric columns in **Liking Variables**, one per consumer.

Each row corresponds to a **stimulus**. The two coordinates describe its position in the 2D space, and the **Liking Variables** contain the consumers' scores for that same stimulus.

Before running the analysis in jamovi, check in the **Data** tab that:

- **X-axis** and **Y-axis** are two different numeric variables and both vary across stimuli;
- all **Liking Variables** are numeric and are different from the coordinate variables;
- at least **two** liking variables are selected;
- if **Stimuli Labels** is used, labels are non-missing and unique.

SEDA requires at least **4 stimuli**, but the minimum then depends on the selected model: 4 for **Vector**, 5 for **Circular**, 6 for **Elliptical**, and 7 for **Quadratic**. Consumer classification requires at least **3 consumers**. Isolated missing liking values can be imputed internally from product and consumer effects; however, an entirely missing consumer, a stimulus with no liking scores, or a consumer with no liking variation is rejected.

8.4 Interface guide

The analysis appears in **SEDA > Hedonic Data > Preference Mapping**. The interface contains four variable drop zones, the **Instructions and R Code** section, **Preference Mapping Options**, a collapsible **Consumer Classification** section, and **Graphic Options**.

8.4.1 Variables to complete

8.4.1.1 Stimuli Labels

Role: provides the names displayed for stimuli on the maps. **Impact on the computation:** none on the fitted models. **Practical tip:** use short, complete, and unique labels. If no label variable is supplied, SEDA uses row numbers.

8.4.1.2 X-axis

Role: provides the horizontal coordinate of the stimuli in the representation space. **Impact on the computation:** directly enters the preference models. **Practical tip:** typically use the first component saved from a PCA performed in MEDA.

8.4.1.3 Y-axis

Role: provides the vertical coordinate of the stimuli. **Impact on the computation:** directly enters the models and must come from the same space as **X-axis**. **Practical tip:** typically use the second component from the same analysis.

8.4.1.4 Liking Variables

Role: contains individual liking scores, one column per consumer. **Impact on the computation:** one model is fitted to each column; if classification is enabled, the same profiles are also used to form consumer segments. **Practical tip:** use individual scores rather than product means.

8.4.2 Instructions and R Code

8.4.2.1 Read me before running

Role: displays integrated help in the results. **Impact on the computation:** none. **Practical tip:** keep it checked during the first uses.

8.4.2.2 R code

Role: displays the R script corresponding to the current settings. **Impact on the computation:** none. **Practical tip:** check it when you want to reproduce the analysis in the **Rj Editor**.

8.4.3 Preference Mapping Options

8.4.3.1 Regression model

Role: selects the response-surface form fitted for each consumer.

Model	Structure	Minimum stimuli
Quadratic (default)	Linear + quadratic + interaction	7
Vector	Linear	4
Circular	Linear + common radial curvature	5

Model	Structure	Minimum stimuli
Elliptical	Linear + separate quadratic terms, no interaction	6

Practical tip: start with **Quadratic** when the number and geometry of products support it. With a small product set, a simpler model is often easier to defend.

8.4.3.2 Preference threshold (SD)

Role: sets the standardised level that predicted preference must exceed for a location to be counted as favourable for one consumer. **Impact on the computation:** directly changes the percentage displayed on the map. **Practical tip:** start with **0**. Positive values identify more selective high-preference regions; negative values broaden favourable regions.

8.4.4 Consumer Classification (*collapsed by default*)

8.4.4.1 Classification of consumers

Role: enables segmentation of consumers from their centred liking profiles. **Impact on the computation:** adds cluster tables, the dendrogram, and, when requested, segment-specific preference maps. Disabled by default.

8.4.4.2 Number of clusters

Role: sets the number of groups. **Impact on the computation:** **-1** lets SEDA choose the cut automatically; an integer of at least 2 imposes the number of segments and must remain below the number of consumers.

8.4.4.3 Cluster dendrogram

Role: displays the Ward clustering dendrogram. **Impact on the computation:** none; it visualises the hierarchy used to form the groups. Enabled by default when classification is used.

8.4.4.4 Preference maps by cluster

Role: produces a separate preference map for each consumer cluster. **Impact on the computation:** each map is recomputed using only the consumers in that group while keeping the same coordinates, model, and threshold as the overall map. Enabled by default when classification is used.

8.4.5 Graphic Options (*collapsed by default*)

8.4.5.1 Color for low preference areas

Role: sets the color of low-preference regions. **Impact on the computation:** none. The default is **Blue**.

8.4.5.2 Color for high preference areas

Role: sets the color of high-preference regions. **Impact on the computation:** none. The default is **Red**.

8.5 Step-by-step procedure

1. Open your dataset in jamovi.
2. If you do not yet have coordinates for the 2D space, construct them in **MEDA** with a **PCA**, then save at least two **Coordinates**.
3. Check that **X-axis**, **Y-axis**, and all **Liking Variables** are numeric.
4. Open **SEDA > Hedonic Data > Preference Mapping**.
5. Place the stimulus-name variable in **Stimuli Labels** if available.
6. Place two coordinates from the same representation space in **X-axis** and **Y-axis**.
7. Place the individual consumer scores in **Liking Variables**.
8. Start with **Regression model = Quadratic** if you have at least 7 products and sufficient product-space geometry; otherwise use a simpler model.
9. Leave **Preference threshold (SD) = 0** for a first reading.
10. Read **Overall Preference Map** first.
11. To investigate preference heterogeneity, enable **Classification of consumers** and start with **Number of clusters = -1**.
12. Read **Consumer Cluster Sizes** and **Consumer Cluster Membership**, followed by the dendrogram and **Preference Maps by Consumer Cluster**.
13. Check **R code** if you want to reproduce the exact settings in the Rj Editor.

8.6 Interpretation of results

8.6.1 Overall Preference Map — main graph

The map shows the space supplied by **X-axis** and **Y-axis**, overlaid with the **% of consumers** surface.

Element	Meaning
Areas toward Color for high preference areas	Regions favourable for a high proportion of consumers according to the selected threshold
Areas toward Color for low preference areas	Regions favourable for a low proportion of consumers
Intermediate areas	Intermediate proportions of favourable consumers
Contour lines	Mark selected consumer percentages from 20% up to 95%
Squares with labels	Positions of the tested stimuli in the selected space
Subtitle	Reports the Preference threshold (SD) used

With the default threshold **0**, a high region means that a large proportion of consumers have predicted preference above their own average predicted level there. The map therefore represents neither raw mean liking nor a probability in the statistical sense.

Point of vigilance: the surface is modelled over a grid covering the plane. Regions far from observed stimuli involve more extrapolation and should be interpreted cautiously.

Point of vigilance: a visually smooth surface does not guarantee robustness. The number of stimuli, the selected model, and consumer diversity determine how much information is available.

8.6.2 Consumer Cluster Sizes (*if Classification of consumers is enabled*)

This table gives the **number** and **percentage** of consumers in each cluster. It immediately identifies dominant segments and very small groups, which require more cautious interpretation.

8.6.3 Consumer Cluster Membership *(if Classification of consumers is enabled)*

This table gives the cluster assigned to every consumer. It can be used to reconnect segments with the original liking columns or to characterise the groups in a complementary analysis.

8.6.4 Consumer Cluster Dendrogram *(optional)*

The dendrogram displays the Ward hierarchical clustering computed from centred liking profiles. Consumers that are close in this hierarchy have similar patterns of preference across products, largely independently of their average rating level.

8.6.5 Preference Maps by Consumer Cluster *(optional)*

A preference map is recomputed for every cluster using **the same product space, Regression model, and Preference threshold (SD)** as the overall map. Differences between maps can therefore be interpreted as heterogeneity in preference structure between segments, rather than as changes in the factorial reference space. On a cluster-specific map, the **% of consumers** is calculated within that cluster: 100% means all consumers in that segment, not all consumers in the full panel.

8.7 Example: the `senso_hedo_cocktail` dataset

The `senso_hedo_cocktail` dataset contains the sensory evaluations of 16 cocktails described by 13 attributes, as well as individual liking scores from **100 consumers**. It is used to illustrate the construction of a **preference map**: the sensory space is first constructed by PCA in MEDA, then the hedonic scores are projected into it through this function. The numerical values and graphics discussed below come directly from the analysis results. To load this dataset, open jamovi, click the icon in the top left corner, the three stacked lines, then **Open** → **Data Library** → **SEDA**.

8.7.1 Example settings

To reproduce the main example and also display consumer segmentation, use the following settings.

Option	Value
Stimuli Labels	variable identifying the cocktails
X-axis	Dim. 1 (saved from the PCA)
Y-axis	Dim. 2 (saved from the PCA)

Option	Value
Liking Variables	all hedonic columns (O to DJ)
R code	checked
Regression model	Quadratic
Preference threshold (SD)	0
Classification of consumers	checked
Number of clusters	-1
Cluster dendrogram	checked
Preference maps by cluster	checked
Color for low preference areas	Blue
Color for high preference areas	Red

8.7.2 Step 1 — Construct the sensory space in MEDA (PCA)

In the **MEDA** module, run a PCA on the 13 sensory attributes with the following settings:

Option	Value
Active Variables	the 13 sensory attributes
Scale to unit variance	Checked
Number of components (Save)	2
Coordinates (Save)	Checked

The PCA produces a two-dimensional space: **Dim. 1 explains 53.25%** of the variance and **Dim. 2 explains 24.13%**, that is, **77.37% cumulative**. The 1/2 plane therefore summarises the sensory profiles of the cocktails well.

The automatic description of the dimensions confirms the interpretation of the axes:

- **Dim. 1** opposes *acidity* (correlation 0.93), *bitterness* (0.88), *odor.orange* (0.75), and *intensity* (0.73) on one side, to *odor.banana* (-0.93), *sweet* (-0.93), *thickness* (-0.88), and *pulp* (-0.70) on the other. This is a clear opposition between acidic and intense cocktails vs sweet, thick, banana-like cocktails.
- **Dim. 2** is more strongly structured by *odor.lemon* (0.77), *odor.mango* (0.69), *persistence* (0.66), and *color.intensity* (0.58), with *odor.orange* (-0.61) on the negative side.

On the individual map, the cocktails are widely distributed across the plane. **386** is located in the upper right (high on Dim. 2), **131** at the bottom centre, **447** and **312** on the left (sweet, thick), and **194**, **233**, and **245** on the right (acidic, intense). Once the PCA has been run, the **Dim. 1** and **Dim. 2** columns automatically appear in the jamovi data sheet.

8.7.3 Step 2 — Construct the preference map in SEDA

Return to **SEDA > Hedonic Data > Preference Mapping** and use the saved coordinates as **X-axis** and **Y-axis**. Select the hedonic columns in **Liking Variables**, keep **Regression model = Quadratic**, and use **Preference threshold (SD) = 0**.

To reproduce the segment-based reading as well, enable **Classification of consumers**, leave **Number of clusters = -1**, and keep **Cluster dendrogram** and **Preference maps by cluster** checked. Enabling classification does not change the overall map; it adds complementary results about consumer heterogeneity.

8.7.4 Analysis of the *Overall Preference Map* graph

The resulting map superimposes the preference surface onto the position of the 16 cocktails in the sensory space. The dark red area — corresponding to more than 80–90% favourable consumers — is located **on the left of the plane**, in the region with negative coordinates on Dim. 1, where the cocktails **182**, **444**, **343**, and **447** are located. These cocktails, characterised by sweeter, thicker, banana-like profiles, lie in a high-preference region.

Conversely, the blue area — corresponding to fewer than 20% favourable consumers — is located **on the right of the plane**, where the cocktails **194**, **233**, and **245** are located, characterised by acidic and intense profiles.

The cocktails **131** (bottom centre) and **386** (upper right) are located in intermediate to low-preference areas (light blue, 10–25%).

8.7.5 Reading the consumer classification

With **Number of clusters = -1**, the Ward classification applied to the centred liking profiles retains **2 consumer clusters** in this example:

- **Cluster 1: 62 consumers** (62.0%). The products with the highest relative liking in the centred cluster profile are **182**, **447**, **472**.
- **Cluster 2: 38 consumers** (38.0%). The products with the highest relative liking in the centred cluster profile are **343**, **312**, **182**.

Product **182** appears among the highest relative liking values in both clusters, whereas **447/472** distinguish Cluster 1 and **343/312** distinguish Cluster 2 among the observed cocktails. This gives a concrete first indication of both shared and segment-specific preference patterns.

These short descriptions are based on the **mean centred liking profile within each cluster**. They identify which tested products most distinguish the relative preferences of a segment; they do not, by themselves, assign a sensory meaning to that segment.

Use **Consumer Cluster Sizes** to assess the weight of each segment and **Consumer Cluster Membership** to identify the consumers composing it. The **Consumer Cluster Dendrogram** shows the hierarchy underlying the partition.

The **Preference Maps by Consumer Cluster** should then be compared in the common sensory space. Because every cluster map uses exactly the same product coordinates, regression model, and preference threshold as the overall map, differences between maps reflect heterogeneity in predicted preference patterns rather than a change of reference space. Very small clusters should nevertheless be interpreted cautiously.

8.7.6 Synthetic interpretation

The sensory space structured by the PCA, based on the acidic-intense vs sweet-thick opposition, is directly readable on the preference map. The highest-preference region lies on the sweet-thick side, whereas acidic and intense profiles are less appreciated. No cocktail appears to occupy the centre of this highest-preference region, which suggests a **direction for formulation improvement** rather than an already optimal product.

What we take away from this example: preference mapping directly links the sensory space and hedonic preferences, making it possible to identify a favourable region of the product space even when no existing product occupies its centre.

8.8 From jamovi to the Rj Editor

Once the **Rj Editor** has been configured with **System R** as described in Chapter 1, check **R code** to display the script corresponding to the analysis. The dataset open in jamovi is available as **data**.

The generated code reconstructs the two coordinates and liking columns, fits a separate response surface for each consumer, standardises predictions over the grid, and applies **Preference threshold (SD)** before computing the **% of consumers**. It therefore reproduces the logic of the map explicitly rather than treating the result as a simple mean liking score.

If **Classification of consumers** is enabled, the script also centres the liking profiles, computes distances between consumers, performs Ward clustering with `hclust(..., method = "ward.D2")`, uses `cutree()` to form the groups, and then recomputes a map for the consumers in each cluster. The generated code therefore reproduces both the overall map and the full segmentation in the Rj Editor.

8.9 Practical tips

- Always check that **X-axis** and **Y-axis** come from the same representation space and refer to the same stimuli as the liking scores.
- Use **individual scores** in **Liking Variables**, one column per consumer.
- Start with **Preference threshold (SD) = 0**; then vary it to assess whether high-preference regions remain stable.
- Match **Regression model** complexity to the number of products: the minima are 4, 5, 6, and 7 products for Vector, Circular, Elliptical, and Quadratic respectively.
- For external preference mapping, interpret the sensory axes before translating a favourable region into a formulation direction.
- If you enable **Classification of consumers**, inspect cluster sizes first and be cautious with very small segments.
- Compare cluster maps in the same reference space: differences reflect different preference structures, not different factorial spaces.
- Check **R code** when you want a reproducible record of the settings and calculations.

8.10 In summary

Preference mapping links a **two-dimensional stimulus space** to **individual liking scores**. SEDA fits a preference surface for each consumer, standardises the predictions, and represents the proportion of consumers exceeding **Preference threshold (SD)** in each region of the plane. The overall map can be complemented by **Ward clustering of centred liking profiles** and by cluster-specific preference maps. The analysis can therefore distinguish a broadly shared preference region from an overall pattern that hides several consumer segments, while keeping the same product space for all comparisons.